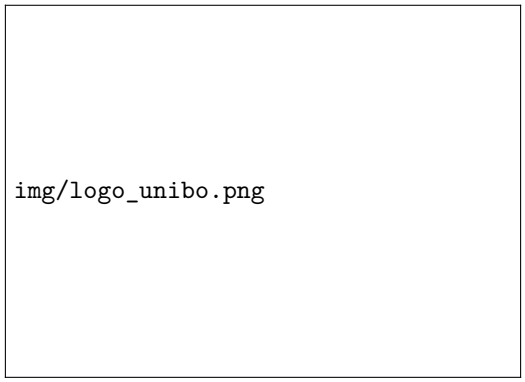


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Theoretical Physics

Quantum Many Body Theory

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Introduction

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1 | Weakly Interacting Bose Gas

1.1 | The Weak Interaction Assumption

The weakly interacting Bose gas serves as a foundational paradigm in quantum many-body theory. We consider a system of N identical bosons of mass m confined to a box of volume $V = L^3$ with periodic boundary conditions.

The interparticle interaction is characterized by a two-body potential $U(\mathbf{r})$, which we assume to be weak. The constituents of this gas obey Bose-Einstein statistics and are treated as non-relativistic elementary particles, meaning that their internal degrees of freedom are not activated at the relevant energy scales, and the total particle number is strictly fixed (precluding, for example, a gas of photons).

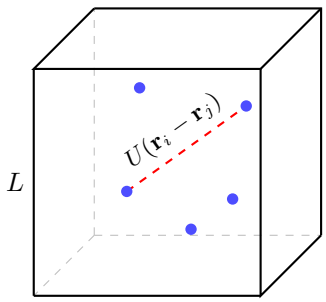


Figure 1.1: Schematic representation of a system of N interacting bosons confined within a cubic box of volume $V = L^3$. The periodic boundary conditions simulate a bulk system. The dashed red line illustrates the two-body interaction potential $U(\mathbf{r}_i - \mathbf{r}_j)$ acting strictly between pairs of particles.

Typical experimental realizations involve dilute gases of bosonic atoms, meaning atoms possessing an even total number of constituent fermions (protons, neutrons, and electrons), such as Sodium (^{23}Na), Rubidium (^{87}Rb), Potassium (^{41}K), Cesium (^{133}Ce), and Lithium (^{7}Li). Historically, this theoretical framework was first motivated by efforts to explain the superfluid behavior of Helium-4 below 2.17K, despite the fact that liquid ^{4}He actually resides in a strongly interacting regime.

The Hamiltonian governing this system of N interacting identical bosons is given in first quantization by

$$\hat{H} = \sum_{i=1}^{N} \frac{\hat{\mathbf{p}}_i^2}{2m} + \frac{1}{2} \sum_{\substack{i,j \\ i \neq j}} U(\hat{\mathbf{r}}_i - \hat{\mathbf{r}}_j),$$

where $\hat{\mathbf{p}}_i$ is the momentum operator of the i-th particle, and $U(\hat{\mathbf{r}}_i - \hat{\mathbf{r}}_j)$ represents the interaction energy between the i-th and j-th particles.

For a system comprising a macroscopic number of identical bosons, it is highly advantageous to

transition to the second quantization formalism.[1] We introduce bosonic creation and annihilation operators associated with the momentum eigenfunctions of a single particle. For a box with periodic boundary conditions, these single-particle states are quantized plane waves:

$$\phi_{\mathbf{k}}(\mathbf{r}) = \frac{e^{i\mathbf{k}\cdot\mathbf{r}}}{\sqrt{V}},$$

where the allowed wave vectors $\mathbf{k}$ take the discrete values

$$\mathbf{k} = \frac{\pi}{L}(n_x, n_y, n_z), \quad n_x, n_y, n_z \in \{0, \pm 1, \pm 2, \dots\}.$$

In this momentum basis, the many-body Hamiltonian becomes

$$\hat{H} = \sum_{\mathbf{k}} \epsilon_k \hat{a}^\dagger_{\mathbf{k}} \hat{a}_{\mathbf{k}} + \frac{1}{2V} \sum_{\mathbf{k},\mathbf{k}',\mathbf{q}} U(\mathbf{q}) \hat{a}^\dagger_{\mathbf{k}+\mathbf{q}} \hat{a}^\dagger_{\mathbf{k}'-\mathbf{q}} \hat{a}_{\mathbf{k}'} \hat{a}_{\mathbf{k}}. \tag{1.1.1}$$

Here, ϵ_k denotes the non-interacting energy associated with the wave vector $\mathbf{k}$ (which depends exclusively on its magnitude k), given by the free-particle dispersion

$$\epsilon_k = \frac{\hbar^2 k^2}{2m},$$

and $U(\mathbf{q})$ is the Fourier transform of the real-space two-body potential $U(\hat{\mathbf{r}}_1 - \hat{\mathbf{r}}_2)$. The operators $\hat{a}^\dagger_{\mathbf{k}}$ and $\hat{a}_{\mathbf{k}}$ represent the creation and destruction operators for the corresponding momentum eigenstates.

In the idealized limit where interactions are completely absent, the ground state of the system is trivial: all N bosons condense into the lowest-energy single-particle state featuring zero momentum ($\mathbf{k} = 0$). This macroscopic formation of a condensate is mathematically expressed by acting N times with the zero-momentum creation operator $\hat{a}^\dagger_0$ on the vacuum state $|0\rangle$ (incorporating the standard $1/\sqrt{N!}$ normalization factor for formal completeness):

$$\left|\psi_0^{(0)}\right\rangle = \frac{(\hat{a}^\dagger_0)^N}{\sqrt{N!}} |0\rangle \,. \tag{1.1.2}$$

Remark. *This clustering phenomenon stands in stark contrast to the behavior of a gas of fermions. Governed by the Pauli exclusion principle, fermions are strictly forbidden from occupying identical quantum states. Consequently, even at zero temperature, non-interacting fermions must sequentially fill all available momentum states up to a maximum threshold known as the Fermi energy, precluding any analogous zero-momentum condensation.*

We now introduce the principal assumption of our approach: the interparticle interaction is sufficiently weak. Given this, we anticipate that the true, interacting ground state remains a modest perturbation of the unperturbed condensate state $\left|\psi_0^{(0)}\right\rangle$. Therefore, even in the presence of interactions, the vast majority of the bosons will remain in the zero-momentum single-particle state. This physical condition is formalized by the inequality

$$\frac{N - N_0}{N} \ll 1, \tag{1.1.3}$$

where $N_0 = \langle \hat{a}^\dagger_0 \hat{a}_0 \rangle$ defines the thermodynamic average number of bosons residing in the condensate. The quantity $N - N_0$ thus counts the number of depleted bosons that have been excited out of the condensate into states with $\mathbf{k} \neq 0$. Due to the weakness of the interaction, these non-zero

[1] For a very clear introduction to the second quantization formalism for many-particle systems, see: H. Tasaki, https://arxiv.org/pdf/1812.10732v5.

momentum states are only microscopically occupied, meaning their occupation numbers scale as $\langle \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} \rangle = \mathcal{O}(1)$. Crucially, the population of these excited states does not scale proportionally with the macroscopic total particle number N.

We leverage this assumption to implement a fundamental approximation on the interaction term of the Hamiltonian (1.1.1). Because operators acting on the condensate ($\hat{a}_0$ and $\hat{a}_0^\dagger$) scale macroscopically with $\sqrt{N_0} \approx \sqrt{N}$, we can safely neglect all interaction terms containing more than two operators equipped with non-zero momentum ($\hat{a}_{\mathbf{k}}$ or $\hat{a}_{\mathbf{k}}^\dagger$ for $\mathbf{k} \neq 0$). (Note that terms containing exactly three zero-momentum operators are inherently forbidden by momentum conservation requirements).

The most substantial and relevant term that survives this truncation is the one containing exclusively operators acting on the condensate ($\mathbf{q} = 0$ and $\mathbf{k} = \mathbf{k}' = 0$):

$$\frac{1}{2V} U(\mathbf{q} = 0) \hat{a}_0^\dagger \hat{a}_0^\dagger \hat{a}_0 \hat{a}_0 = \frac{1}{2V} U(\mathbf{q} = 0)(\hat{N}_0^2 - \hat{N}_0).$$

In the thermodynamic limit, $\hat{N}_0$ diverges linearly with N. Thus, we can completely disregard the trailing $\hat{N}_0$ term in the above expression, as it scales infinitely slower than the dominant quadratic term $\hat{N}_0^2 \sim N^2$.

To systematically express this leading-order contribution in terms of the conserved total particle number, we recognize that the number of bosons in the ground state can be written as the difference between the total number operator $\hat{N}$ and the sum of occupation numbers for all excited states:

$$\hat{N}_0 = \hat{N} - \sum_{\mathbf{k} \neq 0} \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}},$$

where the primed summation specifically denotes a sum over all allowed momenta excluding zero. Letting $U_0 \equiv U(\mathbf{q} = 0)$, we substitute this relation back into our dominant interaction term. Expanding the square and purposefully discarding the resulting quartic term (which contains four out-of-condensate operators and is therefore negligible under our approximation), we obtain

$$\frac{1}{2V} U_0 (\hat{N}_0)^2 = \frac{1}{2V} U_0 \left(\hat{N} - \sum_{\mathbf{k} \neq 0} \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} \right)^2 \approx \frac{1}{2V} U_0 \left(\hat{N}^2 - 2\hat{N} \sum_{\mathbf{k} \neq 0} \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} \right).$$

Through this procedure, we successfully reduce the pure zero-momentum interaction into a quadratic form that couples the macroscopic total number N with the single-particle excitations.

Quadratic Interaction Terms and the Short-Range Assumption

Besides the dominant term where all operators are in the condensate state, we must evaluate the terms from the interaction Hamiltonian that contain exactly two operators with momentum different from zero. Before explicitly listing them, we introduce a crucial physical approximation: we replace the momentum-dependent interaction potential $U(\mathbf{q})$ with a simple constant, $U(0)$.

This approximation is mathematically justified if the real-space potential is effectively "short-ranged". For the Fourier transform $U(\mathbf{q}) = \int \mathrm{d}^3\mathbf{r} U(\mathbf{r}) e^{-i\mathbf{q}\cdot\mathbf{r}}$ to be well-behaved and approximately constant for small momenta, its zero-momentum limit $U(0) = \int \mathrm{d}^3\mathbf{r} U(\mathbf{r})$ must be finite. In three spatial dimensions, the spherical integration measure introduces a factor of r^2. Consequently, for the integral $\int dr r^2 U(r)$ to converge at large interparticle distances, the potential $U(r)$ must decay strictly faster than $1/r^3$.

This convergence criterion dictates which physical systems can be described by this simplified model. In dilute atomic gases, van der Waals forces serve as an excellent example of a valid short-ranged potential. In the standard Lennard-Jones model, the long-range attractive tail scales as $1/r^6$. Since $6 > 3$, it comfortably satisfies the convergence requirement. In contrast, dipole-dipole interactions scale exactly as $1/r^3$. This is a tricky marginal case where the spatial integral diverges logarithmically, meaning dipole interactions are mathematically long-ranged in 3D and cannot be treated with this simple approximation. Assuming our potential decays fast enough, we can safely set $U(\mathbf{q}) \approx U(\mathbf{q} = 0) \equiv U_0$, making all subsequent calculations significantly simpler.

Remark. *As an analogous example of effective short-range interactions from high-energy physics, the strong nuclear force is also effectively short-ranged (decaying exponentially, much faster than $1/r^3$) because its mediators, pions, are massive.*

With the assumption $U(\mathbf{q}) \approx U_0$ established, we retain the following six terms containing exactly two non-zero momentum operators:

1. If $\mathbf{k} + \mathbf{q} = 0$ and $\mathbf{k}' - \mathbf{q} = 0$, we have that $\mathbf{k} = -\mathbf{q}$ and $\mathbf{k}' = \mathbf{q}$, and so we remain with

$$\frac{1}{2V} U_0 a_0^\dagger a_0^\dagger {\sum_{\mathbf{q}}}' a_{\mathbf{q}} a_{-\mathbf{q}},$$

   where the sum $\sum'_{\mathbf{q}}$ is over all possible momenta $\mathbf{q} \neq 0$ because the zero case has been already taken into account in the previous purely macroscopic term.

2. If $\mathbf{k}+\mathbf{q} = 0$ and $\mathbf{k}' = 0$, using again the short-range assumption, the term takes the following form:

$$\frac{1}{2V} U_0 {\sum_{\mathbf{q}}}' a_0^\dagger a_{-\mathbf{q}}^\dagger a_0 a_{-\mathbf{q}} = \frac{1}{2V} U_0 a_0^\dagger a_0 {\sum_{\mathbf{q}}}' a_{-\mathbf{q}}^\dagger a_{-\mathbf{q}}.$$

3. If $\mathbf{k} + \mathbf{q} = 0$ and $\mathbf{k} = 0$, and so $\mathbf{q} = 0$, we get

$$\frac{1}{2V} U_0 {\sum_{\mathbf{k}'}}' a_0^\dagger a_{\mathbf{k}'}^\dagger a_{\mathbf{k}'} a_0 = \frac{1}{2V} U_0 a_0^\dagger a_0 {\sum_{\mathbf{k}'}}' a_{\mathbf{k}'}^\dagger a_{\mathbf{k}'}.$$

4. If $\mathbf{k}' - \mathbf{q} = 0$ and $\mathbf{k}' = 0$, and so $\mathbf{q} = 0$, we get

$$\frac{1}{2V} U_0 {\sum_{\mathbf{k}}}' a_{\mathbf{k}}^\dagger a_0^\dagger a_0 a_{\mathbf{k}} = \frac{1}{2V} U_0 a_0^\dagger a_0 {\sum_{\mathbf{k}}}' a_{\mathbf{k}}^\dagger a_{\mathbf{k}}.$$

5. If $\mathbf{k}' - \mathbf{q} = 0$ and $\mathbf{k} = 0$, with the short-range assumption the term takes the following form:

$$\frac{1}{2V} U_0 {\sum_{\mathbf{q}}}' a_{\mathbf{q}}^\dagger a_0^\dagger a_{\mathbf{q}} a_0 = \frac{1}{2V} U_0 a_0^\dagger a_0 {\sum_{\mathbf{q}}}' a_{\mathbf{q}}^\dagger a_{\mathbf{q}}.$$

6. If $\mathbf{k}' = 0$ and $\mathbf{k} = 0$, with the short-range assumption the term takes the following form:

$$\frac{1}{2V} U_0 {\sum_{\mathbf{q}}}' a_{\mathbf{q}}^\dagger a_{-\mathbf{q}}^\dagger a_0 a_0 = \frac{1}{2V} U_0 a_0 a_0 {\sum_{\mathbf{q}}}' a_{\mathbf{q}}^\dagger a_{-\mathbf{q}}^\dagger.$$

Condensate Mean-Field Approximation and the Bogoliubov Hamiltonian

A further crucial step is the replacement of the operators a_0 and $a_0^\dagger$ with numbers. Because the non-interacting ground state exhibits macroscopic occupation, its expectation value is $\langle a_0^\dagger a_0 \rangle = N_0 \approx N$. Therefore, we can confidently approximate $a_0 \sim \sqrt{N}$ and $a_0^\dagger \sim \sqrt{N}$ across all the previously derived terms of the approximate Hamiltonian that contain two condensate operators multiplied by two out-of-condensate operators.[2]

This mean-field approximation is formally justified by inspecting the commutator $[a_0, a_0^\dagger] = 1$ and comparing its magnitude to the result of acting with these operators on the non-interacting N-particle ground state $\left|\psi_0^{(0)}(N)\right\rangle$. Specifically, one finds:

$$a_0 \left|\psi_0^{(0)}(N)\right\rangle = \sqrt{N} \left|\psi_0^{(0)}(N-1)\right\rangle \sim \sqrt{N} \left|\psi_0^{(0)}(N)\right\rangle,$$

and

$$a_0^\dagger \left|\psi_0^{(0)}(N)\right\rangle = \sqrt{N+1} \left|\psi_0^{(0)}(N+1)\right\rangle \sim \sqrt{N} \left|\psi_0^{(0)}(N)\right\rangle,$$

where we have safely neglected 1 relative to the macroscopically large number N in these expressions. Because of this scaling, the value of the commutator is entirely negligible, and the operators can be effectively treated as commuting c-numbers.

Furthermore, during this systematic expansion, it is important to recognize why linear terms in $a_{\mathbf{k}}$ or $a_{\mathbf{k}}^\dagger$ do not appear. The terms which cancel out are precisely the ones that contain a single $\hat{a}_0$ or $\hat{a}_0^\dagger$ in the interaction term; if they did not cancel, they would have contributed to an unphysical renormalization of the free term of the Hamiltonian.

By substituting these numerical expectations back into our aggregated interaction terms, the full Hamiltonian becomes:

TODO: expand computations.

$$H = \sum_{\mathbf{k}}{}' (\epsilon_k + nU_0) a_{\mathbf{k}}^\dagger a_{\mathbf{k}} + \frac{nU_0}{2} \sum_{\mathbf{k}}{}' (a_{-\mathbf{k}}^\dagger a_{\mathbf{k}}^\dagger + a_{\mathbf{k}} a_{-\mathbf{k}}) + \frac{1}{2} V n^2 U_0, \tag{1.1.4}$$

where $n = N/V$ is the particle density.

Note that the second sum in the above equation contains the "anomalous" terms $a_{-\mathbf{k}}^\dagger a_{\mathbf{k}}^\dagger$ and $a_{\mathbf{k}} a_{-\mathbf{k}}$. Because the Hamiltonian now pairs creation operators together (and annihilation operators together) rather than consisting purely of standard number operators, it is not diagonal. To resolve this and find the true elementary excitations, one must employ the Bogoliubov transformation to diagonalize the Hamiltonian.

As a final remark, we note that if one were to carry out this derivation without invoking the constant potential approximation $U(\mathbf{q}) \simeq U_0$, the resulting Hamiltonian would naturally retain the momentum dependence of the interaction:

$$H = \sum_{\mathbf{k}}{}' (\epsilon_k + nU(k)) a_{\mathbf{k}}^\dagger a_{\mathbf{k}} + \frac{n}{2} \sum_{\mathbf{k}}{}' U(k) (a_{-\mathbf{k}}^\dagger a_{\mathbf{k}}^\dagger + a_{\mathbf{k}} a_{-\mathbf{k}}) + \frac{1}{2} V n^2 U_0.$$

[2] This can be preatty disturbing, since the expectation value of a single ladder operator is usually zero; but, as we will see later, we are not using eigenstates of number operator, instead superpositions of those (similar to coherent states). Indeed, even if the hamiltonian conserves the number of particle, the ground state description does not really rely on a fixed occupation of bosons. We will stress this point later on.

Summary of Assumptions and Approximations

Before proceeding to diagonalize the Hamiltonian, it is instructive to summarize the hierarchy of physical assumptions and mathematical approximations that have systematically reduced the complex many-body problem to the quadratic Bogoliubov Hamiltonian (1.1.4):

- **Non-Relativistic Elementary Bosons:** We assume the particles are massive, non-relativistic, and constrained to energy scales where internal degrees of freedom are completely frozen. *Consequence:* The total particle number N is strictly conserved (unlike in a photon gas), and the system is well-described by standard Galilean kinetic energy and a static scalar two-body potential.
- **Weak Interaction Hypothesis:** The interparticle potential is assumed to be sufficiently weak such that the true interacting ground state is only a modest perturbation of the ideal Bose-Einstein condensate. *Consequence:* The fraction of depleted particles is vanishingly small ($(N - N_0)/N \ll 1$). This justifies truncating the interaction Hamiltonian to discard all terms containing more than two out-of-condensate operators ($\mathbf{k} \neq 0$).
- **Short-Range Potential:** The two-body interaction $U(\mathbf{r})$ is assumed to decay strictly faster than $1/r^3$ at large interparticle distances. *Consequence:* The Fourier transform of the potential converges and varies slowly at small momenta, allowing us to approximate $U(\mathbf{q}) \approx U(\mathbf{q} = 0) \equiv U_0$. This dramatically simplifies the scattering amplitudes in the dilute gas regime.
- **Macroscopic Condensate Occupation (Mean-Field Limit):** The zero-momentum state is macroscopically occupied ($N_0 \approx N \gg 1$), meaning the fundamental commutator $[\hat{a}_0, \hat{a}_0^\dagger] = 1$ is negligible compared to the absolute particle number. *Consequence:* The macroscopic creation and annihilation operators $\hat{a}_0^\dagger$ and $\hat{a}_0$ can be treated as commuting c-numbers and replaced by $\sqrt{N}$. This breaks the $U(1)$ phase symmetry of the Hamiltonian and reduces the complicated quartic interaction term into a tractable quadratic form featuring anomalous pairing terms.

1.1.1 | Bogoliubov Transformations

We wish to diagonalize the Hamiltonian of the weakly interacting Bose gas by introducing new bosonic annihilation and creation operators, $\alpha_{\mathbf{k}}$ and $\alpha_{\mathbf{k}}^\dagger$, known as Bogoliubov operators. We need to construct a Hamiltonian which, in the end, depends only on a diagonal number operator of the form $\alpha_{\mathbf{k}}^\dagger \alpha_{\mathbf{k}}$, yielding a target structure like:

$$H = \sum_{\mathbf{k}} E_k \alpha_{\mathbf{k}}^\dagger \alpha_{\mathbf{k}} + \text{const},$$

for some energy dispersion E_k. To eliminate the anomalous terms that mix creation and annihilation operators in our previous Hamiltonian, we can perform a Bogoliubov transformation, which is a linear transformation mapping the original particle operators to the new quasiparticle operators. We start with the following standard ansatz for the Bogoliubov transformation:

$$\begin{cases} \alpha_{\mathbf{k}} = u_k a_{\mathbf{k}} + v_k a_{-\mathbf{k}}^\dagger, \\ \alpha_{\mathbf{k}}^\dagger = v_k^* a_{\mathbf{k}}^\dagger + u_k^* a_{-\mathbf{k}}, \end{cases} \tag{1.1.5}$$

for $\mathbf{k} \neq 0$, where u_k and v_k are real coefficients. We now assume that these coefficients are real for simplicity, but they could in principle be complex. These coefficients are defined to depend strictly on the momentum magnitude $k = |\mathbf{k}|$, ensuring that rotational invariance is preserved in momentum space.

Because these newly defined operators must describe valid bosonic excitations, they are strictly required to satisfy the standard canonical bosonic commutation relations. We can verify this by evaluating the commutator between two annihilation operators:

$$\begin{aligned}[\alpha_{\mathbf{k}}, \alpha_{\mathbf{k}'}] &= [u_k a_{\mathbf{k}} + v_k a^\dagger_{-\mathbf{k}}, u_{k'} a_{\mathbf{k}'} + v_{k'} a^\dagger_{-\mathbf{k}'}] \\ &= u_k v_{k'} [a_{\mathbf{k}}, a^\dagger_{-\mathbf{k}'}] + v_k u_{k'} [a^\dagger_{-\mathbf{k}}, a_{\mathbf{k}'}] \\ &= u_k v_{k'} \delta_{\mathbf{k},-\mathbf{k}'} - v_k u_{k'} \delta_{-\mathbf{k},\mathbf{k}'} = 0,\end{aligned}$$

which vanishes as expected. Next, we evaluate the critical commutator between a creation and an annihilation operator:

$$\begin{aligned}[\alpha_{\mathbf{k}}, \alpha^\dagger_{\mathbf{k}'}] &= [u_k a_{\mathbf{k}} + v_k a^\dagger_{-\mathbf{k}}, u_{k'} a^\dagger_{\mathbf{k}'} + v_{k'} a_{-\mathbf{k}'}] \\ &= u_k u_{k'} [a_{\mathbf{k}}, a^\dagger_{\mathbf{k}'}] + v_k v_{k'} [a^\dagger_{-\mathbf{k}}, a_{-\mathbf{k}'}] \\ &= (u_k^2 - v_k^2) \delta_{\mathbf{k},\mathbf{k}'}.\end{aligned}$$

To enforce the correct commutation relations, we must impose a hyperbolic normalization condition on the coefficients:

$$|u_k|^2 - |v_k|^2 = 1. \tag{1.1.6}$$

Provided this condition holds, the transformation linking the old and new operators can be easily inverted. We can express the linear mapping and its Hermitian conjugate in matrix notation:

$$\begin{pmatrix} \alpha_{\mathbf{k}} \\ \alpha^\dagger_{-\mathbf{k}} \end{pmatrix} = \begin{pmatrix} u_k & v_k \\ v_k & u_k \end{pmatrix} \begin{pmatrix} a_{\mathbf{k}} \\ a^\dagger_{-\mathbf{k}} \end{pmatrix}.$$

Because the determinant of the transformation matrix is exactly $u_k^2 - v_k^2 = 1$, the matrix is trivially invertible:

$$\begin{pmatrix} a_{\mathbf{k}} \\ a^\dagger_{-\mathbf{k}} \end{pmatrix} = \begin{pmatrix} u_k & -v_k \\ -v_k & u_k \end{pmatrix} \begin{pmatrix} \alpha_{\mathbf{k}} \\ \alpha^\dagger_{-\mathbf{k}} \end{pmatrix},$$

which explicitly yields the inverted transformation:

$$\begin{cases} a_{\mathbf{k}} = u_k \alpha_{\mathbf{k}} - v_k \alpha^\dagger_{-\mathbf{k}}, \\ a^\dagger_{\mathbf{k}} = u_k \alpha^\dagger_{\mathbf{k}} - v_k \alpha_{-\mathbf{k}}. \end{cases}$$

We now proceed with the formal diagonalization of the Hamiltonian by expressing it entirely in terms of the new Bogoliubov operators:

$$\begin{aligned}H = &\sum_{\mathbf{k}}{}' \left[(\epsilon_k + nU_0) v_k^2 - nU_0 u_k v_k \right] + \frac{1}{2} V n^2 U_0 \\ &+ \sum_{\mathbf{k}}{}' \left[(\epsilon_k + nU_0)(u_k^2 + v_k^2) - 2nU_0 u_k v_k \right] \alpha^\dagger_{\mathbf{k}} \alpha_{\mathbf{k}} \\ &+ \sum_{\mathbf{k}}{}' \left[\frac{nU_0}{2}(u_k^2 + v_k^2) - u_k v_k (\epsilon_k + nU_0) \right] (\alpha^\dagger_{\mathbf{k}} \alpha^\dagger_{-\mathbf{k}} + \alpha_{-\mathbf{k}} \alpha_{\mathbf{k}}).\end{aligned}$$

It may initially appear that we have not made significant progress, as the final line still contains non-diagonal anomalous terms. However, we now have the mathematical freedom to choose arbitrary coefficients that eliminate these specific terms. By selecting an appropriate v_k (which

simultaneously determines u_k via the constraint $u_k^2 = 1 + v_k^2$), we can force the bracketed prefactor of the off-diagonal terms to vanish identically:

$$\frac{nU_0}{2}(u_k^2 + v_k^2) - u_k v_k(\epsilon_k + nU_0) = 0.$$

Through simple algebraic manipulation and applying our normalization condition, this requirement simplifies to a second-order polynomial equation for v_k^2:

$$4E_k^2 v_k^4 + 4E_k^2 v_k^2 - (nU_0)^2 = 0,$$

where we have suggestively defined the factor E_k^2 as:

$$E_k^2 = (\epsilon_k + nU_0)^2 - (nU_0)^2. \tag{1.1.7}$$

Solving this equation for v_k^2 directly yields the functional form of the transformation coefficients:

$$v_k^2 = \frac{1}{2}\left(\frac{\epsilon_k + nU_0}{E_k} - 1\right),$$

and consequently,

$$u_k^2 = \frac{1}{2}\left(\frac{\epsilon_k + nU_0}{E_k} + 1\right),$$

which implies for the cross-term:

$$u_k v_k = \frac{nU_0}{2E_k}.$$

By inserting these rigorously derived coefficients back into our expanded operator expression, the off-diagonal terms strictly cancel, yielding the final form of the diagonalized Hamiltonian:

$$H = \sum_{\mathbf{k}}{}' E_k \alpha_{\mathbf{k}}^\dagger \alpha_{\mathbf{k}} + \mathcal{E}_0, \tag{1.1.8}$$

where $\mathcal{E}_0$ is a macroscopic constant energy shift defined by:

$$\mathcal{E}_0 = \frac{1}{2}Vn^2U_0 - \frac{1}{2}\sum_{\mathbf{k}}{}'(\epsilon_k + nU_0 - E_k).$$

The quantity $E_k = [(\epsilon_k + nU_0)^2 - (nU_0)^2]^{1/2} > 0$ represents the fundamental quasiparticle excitation spectrum of the system.

The Non-Interacting Limit

In the non-interacting limit, characterized by a vanishing interaction strength $U_0 \to 0$, it is instructive to verify that our diagonalized theory smoothly recovers the foundational physics of the ideal Bose gas.

First, we examine the Bogoliubov quasiparticle excitation spectrum E_k. By taking the limit of the dispersion relation, we find:

$$E_k = \lim_{U_0 \to 0} \sqrt{(\epsilon_k + nU_0)^2 - (nU_0)^2} = \sqrt{\epsilon_k^2} = \epsilon_k.$$

As physically expected, the quasiparticle spectrum trivially reduces to the free-particle kinetic energy $\epsilon_k = \hbar^2 k^2/(2m)$.

Next, we evaluate the asymptotic behavior of the Bogoliubov transformation coefficients v_k^2 and u_k^2. Substituting $E_k \to \epsilon_k$ and $U_0 \to 0$, we obtain:

$$v_k^2 = \lim_{U_0 \to 0} \frac{1}{2} \left(\frac{\epsilon_k + nU_0}{E_k} - 1 \right) = \frac{1}{2}(1-1) = 0,$$

and correspondingly,

$$u_k^2 = \lim_{U_0 \to 0} \frac{1}{2} \left(\frac{\epsilon_k + nU_0}{E_k} + 1 \right) = \frac{1}{2}(1+1) = 1.$$

This implies that $u_k \to 1$ and $v_k \to 0$. Consequently, the Bogoliubov quasiparticle operators $\alpha_{\mathbf{k}} = u_k a_{\mathbf{k}} + v_k a^\dagger_{-\mathbf{k}}$ reduce identically to the bare particle operators $a_{\mathbf{k}}$. In the absence of interactions, the elementary excitations of the system are simply the original non-interacting bosons themselves, and the quantum depletion of the condensate drops to zero.

Finally, we analyze the macroscopic constant energy shift $\mathcal{E}_0$. The purely classical mean-field contribution $\frac{1}{2}Vn^2U_0$ obviously vanishes. For the quantum fluctuation sum, we note that $E_k \approx \epsilon_k + nU_0 - \frac{(nU_0)^2}{2\epsilon_k}$ for small U_0, meaning the summand $(\epsilon_k + nU_0 - E_k) \sim \mathcal{O}(U_0^2)$. Therefore, the entire shift vanishes:

$$\lim_{U_0 \to 0} \mathcal{E}_0 = 0.$$

The ground state energy of the system correctly defaults to zero, recovering the trivial result for an ideal Bose-Einstein condensate where all particles occupy the lowest available energy level with zero kinetic energy.

Ground State

With the Hamiltonian successfully diagonalized, we first seek its ground state, denoted $|\psi_0\rangle$. This state is mathematically defined as the vacuum of the new Bogoliubov annihilation operators $\alpha_{\mathbf{k}}$ for all non-zero momenta:

$$\alpha_{\mathbf{k}}|\psi_0\rangle = 0, \quad \forall \mathbf{k} \neq 0,$$

since any mode added to the vacuum would necessarily increase the energy of the system.

The corresponding energy eigenvalue of this ground state is simply the zero-point constant $\mathcal{E}_0$:

$$H|\psi_0\rangle = \mathcal{E}_0|\psi_0\rangle.$$

By acting with the creation operator $\alpha^\dagger_{\mathbf{k}}$ on this vacuum state, we can construct the entire spectrum of excited states:

$$\frac{(\alpha^\dagger_{\mathbf{k}_1})^{n_{\mathbf{k}_1}}}{\sqrt{n_{\mathbf{k}_1}!}} \frac{(\alpha^\dagger_{\mathbf{k}_2})^{n_{\mathbf{k}_2}}}{\sqrt{n_{\mathbf{k}_2}!}} \cdots \frac{(\alpha^\dagger_{\mathbf{k}_N})^{n_{\mathbf{k}_N}}}{\sqrt{n_{\mathbf{k}_N}!}} |\psi_0\rangle,$$

whose corresponding energy eigenvalues are given by:

$$\mathcal{E}_0 + \sum_{i=1}^{N} n_{\mathbf{k}_i} E_{\mathbf{k}_i},$$

where $n_{\mathbf{k}_i} = \alpha^\dagger_{\mathbf{k}_i} \alpha_{\mathbf{k}_i}$ is the occupation number of the i-th state in the *Bogoliubov quasiparticle basis*.

It is critical to note that the vacuum of these newly defined operators, $|\psi_0\rangle$, is distinct from the vacuum of the original bare particle operators, $a_{\mathbf{k}}$. This distinction becomes physically apparent

when we evaluate the expectation value of the bare particle number operator in the interacting ground state:

$$\begin{aligned}\langle\psi_0|a^\dagger_{\mathbf{k}}a_{\mathbf{k}}|\psi_0\rangle &= \langle\psi_0|(u_k\alpha^\dagger_{\mathbf{k}} - v_k\alpha_{-\mathbf{k}})(u_k\alpha_{\mathbf{k}} - v_k\alpha^\dagger_{-\mathbf{k}})|\psi_0\rangle \\ &= \langle\psi_0|\left[u_k^2\alpha^\dagger_{\mathbf{k}}\alpha_{\mathbf{k}} + v_k^2\alpha_{-\mathbf{k}}\alpha^\dagger_{-\mathbf{k}} - u_kv_k(\alpha^\dagger_{\mathbf{k}}\alpha^\dagger_{-\mathbf{k}} + \alpha_{-\mathbf{k}}\alpha_{\mathbf{k}})\right]|\psi_0\rangle \\ &= v_k^2 \neq 0,\end{aligned}$$

where the final evaluation leverages the commutation relation $[\alpha_{-\mathbf{k}}, \alpha^\dagger_{-\mathbf{k}}] = 1$ and the fact that annihilation operators destroy the vacuum. This previous calculation elegantly demonstrates a profound physical consequence of the theory: *even in the absolute zero-temperature ground state* $|\psi_0\rangle$*, the mean occupation number of states with non-zero momentum is strictly positive.* In terms of the original operators, we would state that $n_{\mathbf{k}} \neq 0$ (where $n_{\mathbf{k}} = a^\dagger_{\mathbf{k}}a_{\mathbf{k}}$) even when $\mathbf{k} \neq 0$. Because of the interactions, a finite fraction of bosons is fundamentally pushed out of the zero-momentum state, leading to a phenomenon known as quantum depletion, where the condensate is only partially occupied even at zero temperature.

1.2 | Condensate Depletion and Excitations

In the previous section, we established that the approximate Hamiltonian of a weakly interacting Bose gas can be rewritten in a diagonal form using the Bogoliubov transformation. The ground state $|\psi_0\rangle$ of this diagonalized Hamiltonian is defined precisely as the vacuum of the new Bogoliubov annihilation operators $\alpha_{\mathbf{k}}$, meaning $\alpha_{\mathbf{k}}|\psi_0\rangle = 0$ for all $\mathbf{k} \neq 0$.

In terms of the original bare particle creation and annihilation operators, we can compute the momentum distribution function in this interacting ground state for both the zero-momentum ($\mathbf{k} = 0$) and non-zero-momentum ($\mathbf{k} \neq 0$) states:

$$\langle\psi_0|a_0^\dagger a_0|\psi_0\rangle = N_0 \sim N,$$

$$\langle\psi_0|a_{\mathbf{k}}^\dagger a_{\mathbf{k}}|\psi_0\rangle = v_k^2 \neq 0, \quad \mathbf{k} \neq 0.$$

While in the non-interacting case all N particles condense into the single-particle state with $\mathbf{k} = 0$, the presence of interactions induces a partial quantum depletion of the condensate even at absolute zero.

We now want to calculate the total number of these non-condensed bosons, denoted as N'. This non-condensate fraction is evaluated by summing the expectation value of the bare number operator over all non-zero momentum states:

$$N' = N - N_0 = \sum_{\mathbf{k}}{}' \langle\psi_0|a_{\mathbf{k}}^\dagger a_{\mathbf{k}}|\psi_0\rangle = \sum_{\mathbf{k}}{}' v_k^2,$$

where the primed sum indicates summation over all $\mathbf{k} \neq 0$. As a consistency check, we recall the explicit form of v_k^2 derived earlier; in the strictly non-interacting limit, this excited occupation correctly vanishes:

$$v_k^2 = \frac{1}{2}\left(\frac{\epsilon_k + nU_0}{E_k} - 1\right) \xrightarrow{U_0 \to 0} 0.$$

However, for interacting bosons, there is a fundamental partial depletion of the condensate even at zero temperature. Let us calculate the total number of non-condensed bosons N'. To evaluate this macroscopic quantity, we transition to the thermodynamic limit—taking $V \to \infty$ and $N \to \infty$ while keeping the density $n = N/V$ constant—which allows us to replace the discrete momentum sum with a continuous integral over k-space:

$$N' = \sum_{\mathbf{k}} v_k^2 \xrightarrow[\text{TD limit}]{N,V\to\infty,\ n \text{ const.}} V \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} v_k^2.$$

Dividing both sides by the volume V, the density of non-condensed bosons n' is given by:

$$n' = \frac{N'}{V} = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} v_k^2. \tag{1.2.1}$$

Before explicitly calculating this integral, we will verify its mathematical convergence in both the infrared (long-wavelength, $k \to 0$) and ultraviolet (short-wavelength, $k \to \infty$) limits. When $k \to 0$, the Bogoliubov excitation energy $E_k \to 0$, which simplifies the occupation number to:

$$v_k^2 \sim \frac{nU_0}{2E_k} \quad (k \to 0).$$

Infrared limit. To analyze exactly how E_k scales towards zero, we substitute the free particle dispersion $\epsilon_k = \hbar^2 k^2/2m$:

$$E_k \sim \sqrt{2nU_0\epsilon_k} = c\hbar k \quad (k \to 0),$$

where we have conveniently defined the parameter c as:

$$c = \sqrt{\frac{nU_0}{m}}. \tag{1.2.2}$$

This quantity c possesses the dimensions of a velocity and physically corresponds to the *speed of sound* propagating through the weakly interacting Bose gas. Combining these small-k expressions, the ground state occupation number diverges near the origin as:

$$v_k^2 \sim \frac{mc}{2\hbar k} \quad (k \to 0).$$

Despite this $1/k$ divergence in the integrand, the integration measure $\mathrm{d}^3\mathbf{k}$ in spherical coordinates introduces a geometric factor of k^2, ensuring that the integral (1.2.1) is safely convergent for $k \to 0$.

Ultraviolet limit. Let us now examine the ultraviolet behavior as $k \to \infty$. By Taylor expanding the square root definition of E_k in the limit where kinetic energy dominates the interaction ($\epsilon_k \gg nU_0$), we have:

$$E_k = \epsilon_k\sqrt{1 + \frac{2nU_0}{\epsilon_k}} \sim \epsilon_k\left[1 + \frac{nU_0}{\epsilon_k} - \frac{1}{2}\left(\frac{nU_0}{\epsilon_k}\right)^2\right].$$

Substituting this expansion back into the full expression for v_k^2, the leading-order terms cancel, leaving a rapid decay:

$$v_k^2 \sim \frac{1}{2}\left[\frac{\epsilon_k + nU_0}{\epsilon_k\left(1 + \frac{nU_0}{\epsilon_k} - \frac{1}{2}\left(\frac{nU_0}{\epsilon_k}\right)^2\right)} - 1\right] \sim \frac{(nU_0)^2}{4\epsilon_k^2} = \left(\frac{mc}{\hbar k}\right)^4.$$

Because the occupation number decays as $1/k^4$ at large momenta, the integral converges absolutely at the $k \to \infty$ upper bound as well.

To formally evaluate the integral, we introduce a natural dimensionless momentum variable $\tilde{k}$ defined as:

$$\tilde{k} = \frac{\hbar k}{mc},$$

such that the occupation number can be elegantly rewritten as a universal function of $\tilde{k}$:

$$v_k^2 = v_{\tilde{k}}^2 = \frac{1}{2}\left(\frac{\tilde{k}^2/2 + 1}{\sqrt{\tilde{k}^4/4 + \tilde{k}^2}} - 1\right).$$

This final expression is considered universal in the sense that all system-specific dependencies, such as the particle density n or the interaction potential strength U_0, have been completely absorbed into the definition of the dimensionless coordinate $\tilde{k}$.

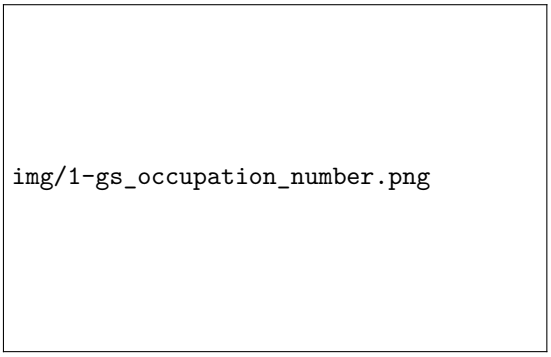


Figure 1.2: Occupation numbers of the ground state $v_{\tilde{k}}^2$. Notice that for small $\tilde{k}$ one has $v_{\tilde{k}}^2 \sim 1/(2\tilde{k})$, while for large $\tilde{k}$ one has $v_{\tilde{k}}^2 \sim 1/\tilde{k}^4$.

Having established its convergence and universality, we can explicitly compute the non-condensed density n' by executing the change of variables from k to $\tilde{k}$ inside the integral:

$$n' = \left(\frac{mc}{\hbar}\right)^3 \int \frac{\mathrm{d}^3\tilde{k}}{(2\pi)^3} \frac{1}{2} \left(\frac{\tilde{k}^2/2 + 1}{\sqrt{\tilde{k}^4/4 + \tilde{k}^2}} - 1 \right) = \left(\frac{mc}{\hbar}\right)^3 \frac{1}{4\pi^2} \int_0^\infty \tilde{k}^2 \left(\frac{\tilde{k}^2/2 + 1}{\sqrt{\tilde{k}^4/4 + \tilde{k}^2}} - 1 \right) \mathrm{d}\tilde{k}.$$

This remaining one-dimensional integral over $\tilde{k}$ can be solved exactly, yielding a factor of $4/3$:

$$\int_0^\infty \tilde{k}^2 \left(\frac{\tilde{k}^2/2 + 1}{\sqrt{\tilde{k}^4/4 + \tilde{k}^2}} - 1 \right) \mathrm{d}\tilde{k} = \frac{4}{3}.$$

Applying this result, the density of depleted bosons elegantly simplifies to:

$$n' = \frac{1}{3\pi^2} \left(\frac{mc}{\hbar}\right)^3.$$

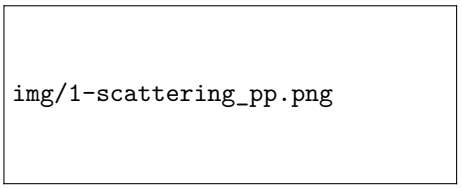
The fundamental fractional depletion of the condensate, given by the ratio $N'/N = n'/n$, is therefore:

$$\frac{N'}{N} = \frac{1}{3\pi^2 n} \left(\frac{mc}{\hbar}\right)^3. \tag{1.2.3}$$

We can thus interpret and manipulate the condensate depletion by operating strictly on the macroscopic density n and the characteristic sound velocity $c = \sqrt{nU_0/m}$. Because c entirely encapsulates the underlying interaction strength U_0, it effectively acts as the single defining free parameter for the weakly interacting system's thermodynamic properties.

Example (Scattering Length). The scattering length a is a measure of the strength of the interaction between two particles. It is defined as the distance at which the wave function of two interacting particles goes to zero. In the context of Bose-Einstein condensates, the scattering length plays a crucial role in determining the properties of the condensate, including its stability and the nature of its excitations.

Figure 1.3: Scattering process with incoming momentum $\mathbf{p}$ and outgoing momentum $\mathbf{p}'$. If the scattering is elastic, momentum conservation ensures that $|\mathbf{p}'| = |\mathbf{p}|$.



To rigorously justify this parameterization, recall that within the Born approximation—which is formally valid for weak interactions—the scattering amplitude $f(\mathbf{p}' \leftarrow \mathbf{p})$ is directly related

to the Fourier transform of the interaction potential $U(\mathbf{q})$ evaluated at the momentum transfer $\mathbf{q} = \mathbf{p}' - \mathbf{p}$:

$$f(\mathbf{p}' \leftarrow \mathbf{p}) = -\frac{m_r}{2\pi\hbar^2} U(\mathbf{q}),$$

where m_r is the reduced mass of the two-body collision. Because we are considering identical bosons of mass m, the reduced mass simplifies to $m_r = m/2$.

At very low collision energies, the momentum transfer approaches zero, and the scattering becomes completely isotropic, entirely losing its angular dependence. In this fundamental low-energy limit, the scattering length a is mathematically defined as the negative asymptotic limit of the scattering amplitude:

$$a = -\lim_{\mathbf{p},\mathbf{p}' \to 0} f(\mathbf{p}' \leftarrow \mathbf{p}).$$

Because the scattering is isotropic, this single parameter is sufficient to completely control both the differential cross-section, $\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = |a|^2$, and the total cross-section, $\sigma_{\text{tot}} = 4\pi a^2$.

By substituting the zero-momentum limit of the Born approximation into this definition, we obtain $a = \frac{m}{4\pi\hbar^2} U_0$. Consequently, for weakly interacting Bose gases, the scattering length can be inverted to define the effective interaction potential U_0 through the relation:

$$U_0 = \frac{4\pi\hbar^2 a}{m}, \tag{1.2.4}$$

where m is the mass of the particles. This relation allows us to express various physical quantities in terms of the scattering length, providing a more intuitive understanding of how interactions affect the condensate.

1.2.1 | Condensate Depletion in Terms of the Gas Parameter

With the effective interaction parameterized strictly by the geometric scattering length a, we can recast our thermodynamic results into a universal form. We first substitute the expression for U_0 into our previous definition of the sound velocity, $c = \sqrt{nU_0/m}$:

$$c = \frac{1}{m}\sqrt{4\pi\hbar^2 n a}.$$

Rearranging these terms to isolate the specific velocity-momentum ratio used in our depletion integral yields:

$$\frac{mc}{\hbar} = 2\sqrt{\pi n a}.$$

We can now insert this expression back into the previously derived equation for the fraction of non-condensed bosons, $\frac{N'}{N} = \frac{1}{3\pi^2 n}\left(\frac{mc}{\hbar}\right)^3$. Performing this substitution directly provides the final analytic result for the condensate depletion:

$$\frac{N'}{N} = \frac{8}{3\sqrt{\pi}}\sqrt{n a^3}. \tag{1.2.5}$$

A profound observation regarding Equation 1.2.5 is the complete absence of the reduced Planck constant, $\hbar$. Because the final result does not contain any $\hbar$ term, there is no mathematical pathway to expand the expression or take a limit to recover a classical or semiclassical analog. This is a direct consequence of the underlying physics: quantum depletion—and Bose-Einstein condensation in general—are purely quantum mechanical effects, fundamentally arising from the zero-point energy and non-commuting nature of the interactions between particles in the condensate.

The quantity N'/N physically represents the fraction of bosons that are pushed out of the condensate ground state. Crucially, Equation 1.2.5 reveals that this depletion depends entirely on a dimensionless quantity called the **gas parameter**, na^3, which scales as the density multiplied by the characteristic scattering volume. Alternatively, we can interpret this parameter as the cube of the ratio between the scattering length a and the average inter-particle distance $n^{-1/3}$.

For the initial Bogoliubov mean-field treatment to be mathematically valid, we explicitly required the depleted fraction N'/N to be small. Consequently, our theory requires the gas parameter to satisfy the strict condition $na^3 \ll 1$. The real physical meaning of this approximation is that it can be satisfied in two distinct ways: the parameter can be small either because the scattering length a is intrinsically small (representing a genuinely weak interaction), or because the particle density n is exceedingly low (representing a dilute gas).

Because of this density dependence, it can be proven that the Bogoliubov theory remains highly accurate even for infinitely strong interactions, provided the gas is sufficiently dilute. Consider, for instance, a hard-sphere core potential parameterized by a radius r_0:

$$V(r) = \begin{cases} +\infty & r < r_0, \\ 0 & r > r_0. \end{cases}$$

For such an impenetrable hard-sphere potential, the scattering length is simply equal to the core radius, $a = r_0$. Even in this highly non-perturbative scenario, the established theory remains perfectly valid as long as $nr_0^3 \ll 1$. This specific condition of ultra-low density is universally realized in experimental systems of ultracold atomic gases.

1.3 | Ground State Energy

Let us now study the macroscopic ground state energy $\mathcal{E}_0$ of the interacting system. Recall that for an ideal, non-interacting Bose gas, the ground state energy is trivially zero because all particles occupy the zero-momentum state without any kinetic or potential cost. However, the introduction of a weak interparticle potential fundamentally alters this scenario. By taking the thermodynamic limit, the discrete sum over the momentum modes transforms into a continuous integral over all possible wave vectors, yielding the energy density:

$$\frac{\mathcal{E}_0}{V} = \frac{1}{2}U_0 n^2 - \frac{1}{2}\int \frac{d^3k}{(2\pi)^3}(\epsilon_k + nU_0 - E_k).$$

An immediate problem arises when we inspect the behavior of this integral at large momenta (the ultraviolet limit). Based on our earlier asymptotic expansion, the Bogoliubov spectrum E_k approaches the free-particle energy with a leading-order correction, meaning the term in the integrand, $\epsilon_k + nU_0 - E_k$, decays asymptotically as $1/k^2$. Because the three-dimensional integration measure d^3k introduces a k^2 factor, the integral diverges linearly.

The physical origin of this unphysical ultraviolet divergence lies in the strict short-range approximation we previously adopted, namely replacing the momentum-dependent potential $U(\mathbf{k})$ with a constant $U(\mathbf{k} = 0) = U_0$ for all $\mathbf{k}$. In reality, only a mathematically pathological Dirac delta potential possesses a Fourier transform that remains constant for arbitrarily large momenta. For any realistic physical potential, the Fourier transform must eventually decay to zero at large wave vectors.

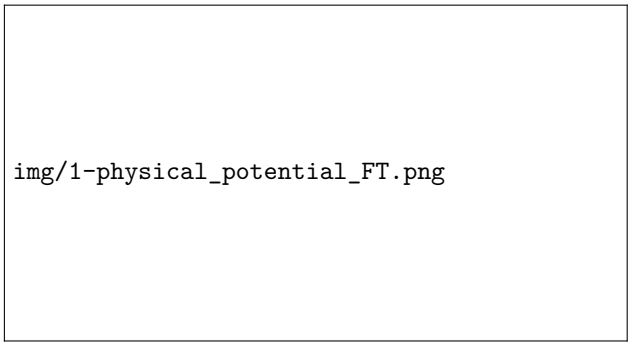


Figure 1.4: Fourier transform of a generic physical potential $U(\mathbf{r})$ in momentum space, $U(\mathbf{k})$. The Fourier transform is a smooth function that decays to zero at large wave vectors. In contrast, the delta-function approximation for the interaction potential is represented by a constant horizontal line, which is valid only for low-momentum scattering processes.

1.3.1 | Renormalization and the Lee-Huang-Yang Correction

Fortunately, the assumption of a strictly constant potential is not strictly necessary for the theory's structural integrity. If one repeats the rigorous diagonalization procedure while retaining the full momentum dependence of the potential, the Bogoliubov transformation coefficients become:

$$v_{\mathbf{k}}^2 = \frac{1}{2}\left(\frac{\epsilon_k + nU(\mathbf{k})}{E_{\mathbf{k}}} - 1\right),$$

and the quasiparticle dispersion takes the generalized form:

$$E_{\mathbf{k}} = \left[(\epsilon_k + nU(\mathbf{k}))^2 - (nU(\mathbf{k}))^2\right]^{1/2}.$$

With these generalized expressions, the corrected ground state energy density becomes:

$$\frac{\mathcal{E}_0}{V} = \frac{1}{2}U_0 n^2 - \frac{1}{2}\int \frac{d^3k}{(2\pi)^3}(\epsilon_k + nU(\mathbf{k}) - E_{\mathbf{k}}). \tag{1.3.1}$$

By evaluating the large momentum limit of this refined integrand, we find:

$$\epsilon_k + nU(\mathbf{k}) - E_{\mathbf{k}} \sim \frac{(nU(\mathbf{k}))^2}{2\epsilon_k}.$$

Therefore, to guarantee the convergence of the energy integral, it is entirely sufficient that the Fourier transform of the potential, $U(\mathbf{k})$, vanishes faster than $1/\sqrt{k}$ at large k. This is a remarkably mild assumption, naturally verified by any real-space potential $U(r)$ that does not diverge faster than $1/r^{5/2}$ as $r \to 0$.

Although this resolves the mathematical divergence, it introduces an interpretational challenge: the ground state energy now explicitly depends on the high-energy microscopic details of the interaction potential $U(\mathbf{k})$ rather than on a single, universally measurable macroscopic parameter like the scattering length a. To eliminate this unpleasant feature and express the energy purely in terms of physical observables, we employ a standard procedure analogous to classical renormalization. We add and subtract the problematic asymptotic ultraviolet behavior from our integral expression:

$$\frac{\mathcal{E}_0}{V} = \frac{U_0 n^2}{2} - \frac{1}{2}\int \frac{d^3k}{(2\pi)^3}\left[\epsilon_k + nU(\mathbf{k}) - E_{\mathbf{k}} - \frac{(nU(\mathbf{k}))^2}{2\epsilon_k} + \frac{(nU(\mathbf{k}))^2}{2\epsilon_k}\right].$$

By splitting the integral and factoring out the density, we obtain:

$$\frac{\mathcal{E}_0}{V} = \frac{n^2}{2}\left(U_0 - \int \frac{d^3k}{(2\pi)^3}\frac{U(\mathbf{k})^2}{2\epsilon_k}\right) - \frac{1}{2}\int \frac{d^3k}{(2\pi)^3}\left[\epsilon_k + nU(\mathbf{k}) - E_{\mathbf{k}} - \frac{(nU(\mathbf{k}))^2}{2\epsilon_k}\right].$$

The bracketed prefactor of the first term can be elegantly linked to the physically observable scattering length a by moving beyond the first-order Born approximation and utilizing the second-order Born approximation for the scattering amplitude $f(\mathbf{p}' \leftarrow \mathbf{p})$:

$$f(\mathbf{p}' \leftarrow \mathbf{p}) = -\frac{m_r}{2\pi\hbar^2}\left[U(\mathbf{p}' - \mathbf{p}) + \int \frac{d^3p''}{(2\pi)^3}\frac{U(\mathbf{p}' - \mathbf{p}'')U(\mathbf{p}'' - \mathbf{p})}{\frac{p^2}{2m_r} - \frac{p''^2}{2m_r} + i0^+}\right],$$

where 0^+ denotes an infinitesimal positive regulator. Recalling the definition of the scattering length in the low-energy limit, the expression simplifies to:

$$a = \frac{m}{4\pi\hbar^2}\left(U_0 - \int \frac{d^3p''}{(2\pi)^3}\frac{U(\mathbf{p}'')^2}{\frac{p''^2}{m} - i0^+}\right).$$

Because the integrand naturally vanishes for $p'' = 0$, the $i0^+$ regulator becomes irrelevant. By changing the integration variable to the wave vector $\mathbf{k}$ and identifying $\epsilon_k = \hbar^2k^2/2m$, we uncover the crucial identity:

$$a = \frac{m}{4\pi\hbar^2}\left(U_0 - \int \frac{d^3k}{(2\pi)^3}\frac{U(\mathbf{k})^2}{2\epsilon_k}\right).$$

By successfully replacing the bare coupling constant U_0 with the physical scattering length a, the ground state energy is renormalized:

$$\frac{\mathcal{E}_0}{V} = \frac{2\pi\hbar^2 n^2 a}{m} - \frac{1}{2}\int \frac{d^3k}{(2\pi)^3}\left(\epsilon_k + nU(\mathbf{k}) - E_{\mathbf{k}} - \frac{(nU(\mathbf{k}))^2}{2\epsilon_k}\right).$$

Because the remaining integral is now manifestly convergent in the ultraviolet regime, we can safely reapply the low-energy approximation $U(\mathbf{k}) \approx U_0$ inside the integral without generating any infinities. The integral then becomes:

$$\int \frac{d^3k}{(2\pi)^3} \left(\epsilon_k + nU_0 - E_k - \frac{(nU_0)^2}{2\epsilon_k} \right).$$

By rewriting this in terms of our previously defined dimensionless momentum variable $\tilde{k}$, the integral scales out as:

$$\left(\frac{mc}{\hbar}\right)^3 mc^2 \frac{1}{2\pi^2} \int_0^\infty \tilde{k}^2 \left(\frac{\tilde{k}^2}{2} + 1 - \sqrt{\frac{\tilde{k}^4}{4} + \tilde{k}^2} - \frac{1}{\tilde{k}^2} \right) d\tilde{k}.$$

This dimensionless integral can be evaluated analytically and yields an exact value of $-32/15$. Inserting this numerical result alongside the definition of the sound velocity c, we arrive at the celebrated final expression for the ground state energy of a dilute Bose gas:

$$\frac{\mathcal{E}_0}{V} = \frac{2\pi\hbar^2 n^2 a}{m} \left[1 + \frac{128}{15\sqrt{\pi}} \sqrt{na^3} \right]. \tag{1.3.2}$$

The leading term in Equation 1.3.2 represents the standard mean-field energy, $\frac{1}{2}n^2U_0$, which precisely matches the expectation value of the Hamiltonian if all quantum field operators were simply replaced by their classical expectation value $\sqrt{n}$. The subsequent correction, proportional to the gas parameter $\sqrt{na^3}$, is the renowned Lee-Huang-Yang (LHY) term. This pivotal term specifically accounts for the zero-point quantum fluctuations of the interacting field around its classical mean-field value.

The macroscopic stability of the gas is dictated by these energy contributions. For a repulsive interaction ($a > 0$), the energy increases quadratically with density, meaning the macroscopic compressibility is positive and the gas will homogenously fill the confining volume to minimize its energy. Conversely, if the effective potential is attractive ($a < 0$), the scattering length is negative, and the energy is minimized by allowing the density to approach infinity. Because bosons are not protected by the Pauli exclusion principle, an attractive Bose gas is mechanically unstable and will spontaneously undergo spatial collapse.

Remark. *The interplay between the mean-field energy and the LHY quantum fluctuations can yield fascinating new phases of matter. In 2015, D. S. Petrov theoretically demonstrated the stabilization of a collapsing binary Bose-Bose mixture. Consider a mixture of two bosonic species with intra-species repulsion ($a_{11}, a_{22} > 0$) but a sufficiently strong inter-species attraction ($a_{12} < 0$) such that the total mean-field compressibility becomes negative, triggering a collapse. Because the attractive mean-field term scales as $\propto n^2$ while the repulsive LHY fluctuation term scales as $\propto n^{5/2}$, the repulsive quantum fluctuations dominate at higher densities. This delicate competition creates an optimal finite density that minimizes the energy, halting the collapse. The resulting state is a self-bound quantum droplet that remains in equilibrium with the vacuum even without external trapping potentials, a phenomenon successfully observed in experiments in 2018.*

Pressure and Speed of Sound

Returning to a single-component Bose gas, we observe that the macroscopic pressure at absolute zero, $T = 0$, is derived from the standard thermodynamic energy-volume relation:

$$P = -\left(\frac{\partial E}{\partial V}\right)_N = -\frac{E}{V} + n\frac{\partial (E/V)}{\partial n}.$$

By substituting the Lee-Huang-Yang expression for the ground state energy density, we obtain the corrected pressure:

$$P = \frac{2\pi\hbar^2 n^2 a}{m}\left[1 + \frac{64}{5\sqrt{\pi}}\sqrt{na^3}\right].$$

From this thermodynamic equation of state, the macroscopic speed of sound, c_s, in the interacting gas is given by the density derivative of the pressure:

$$c_s^2 = \frac{1}{m}\frac{\partial P}{\partial n} = \frac{4\pi\hbar^2 na}{m^2}\left[1 + \frac{16}{\sqrt{\pi}}\sqrt{na^3}\right].$$

We immediately see that to leading order in the gas parameter, the thermodynamic speed of sound c_s agrees exactly with the microscopic velocity c extracted from the linear slope of the single-particle Bogoliubov dispersion E_k in the long-wavelength limit ($k \to 0$) introduced previously. It can be formally proven that the fundamental identity $c = c_s$ should hold exactly in any fully consistent many-body theory[3][4]. However, maintaining strict consistency with the perturbative result for the speed of sound to next-to-leading order requires extending calculations beyond the present Bogoliubov mean-field approximation when computing the quasiparticle spectrum E_k[5].

1.3.2 | Ground State Wave Function

We established that the exact ground state $|\psi_0\rangle$ of the diagonalized effective Hamiltonian must be the vacuum state for the Bogoliubov quasiparticle excitations. Mathematically, this mandates that for all non-zero wave vectors $\mathbf{k} \neq 0$, the annihilation operator yields $\alpha_{\mathbf{k}}|\psi_0\rangle = 0$. If we substitute the defining Bogoliubov transformation—which expresses the quasiparticle operator $\alpha_{\mathbf{k}}$ as a linear combination of the original bare bosonic creation and destruction operators—into this vacuum condition, we obtain:

$$u_k a_{\mathbf{k}}|\psi_0\rangle = -v_k a^\dagger_{-\mathbf{k}}|\psi_0\rangle, \tag{1.3.3}$$

for $\mathbf{k} \neq 0$.

The physical implications of this constraint are profound. On the left-hand side of Equation 1.3.3, the bare annihilation operator $a_{\mathbf{k}}$ acts on the ground state, explicitly removing a boson of momentum $\mathbf{k}$. Simultaneously, on the right-hand side, a creation operator $a^\dagger_{-\mathbf{k}}$ acts on the same ground state, creating a particle with the exact opposite momentum $-\mathbf{k}$. This dynamic equality makes it abundantly clear that the true interacting ground state $|\psi_0\rangle$ cannot be a simple eigenvector of the total particle number operator $\hat{N}$. If it were an eigenstate, the left-hand side of the equation would project onto a subspace with $N - 1$ particles, whereas the right-hand side would project onto a subspace with $N + 1$ particles, creating an irreconcilable contradiction.

[3] C.R. Cabrera, L. Tanzi, J. Sanz, B. Naylor, P. Thomas, P. Cheiney, and L. Tarruell, *Quantum liquid droplets in a mixture of Bose-Einstein condensates*, Science **359**, 301 (2018).
[4] P. C. Hohenberg and P. C. Martin, Annals of Physics **34**, 291 (1965).
[5] S.T. Beliaev, Soviet Phys. JETP **7**, 299 (1958).

Instead, the structural property demanded by Equation 1.3.3 is satisfied by a state heavily featuring coherent superpositions of correlated particle pairs. We propose the following ansatz for the non-condensate portion of the state:

$$\exp\left(-\frac{1}{2}\sum_{\mathbf{p}}{}' \frac{v_p}{u_p} a^\dagger_{\mathbf{p}} a^\dagger_{-\mathbf{p}}\right)|0\rangle.$$

To rigorously verify that this state satisfies our requirement, we act upon it with the operator $u_k a_{\mathbf{k}}$:

$$u_k a_{\mathbf{k}} \exp\left(-\frac{1}{2}\sum_{\mathbf{p}}{}' \frac{v_p}{u_p} a^\dagger_{\mathbf{p}} a^\dagger_{-\mathbf{p}}\right)|0\rangle = u_k \exp\left(-\frac{1}{2}\sum_{\mathbf{p}}{}' \frac{v_p}{u_p} a^\dagger_{\mathbf{p}} a^\dagger_{-\mathbf{p}}\right) a_{\mathbf{k}}|0\rangle$$
$$+u_k \left[a_{\mathbf{k}}, \exp\left(-\frac{1}{2}\sum_{\mathbf{p}}{}' \frac{v_p}{u_p} a^\dagger_{\mathbf{p}} a^\dagger_{-\mathbf{p}}\right)\right]|0\rangle.$$

The first term trivially vanishes because the bare destruction operator $a_{\mathbf{k}}$ acts directly on the absolute vacuum $|0\rangle$. The second term can be explicitly evaluated by expanding the exponential function as a Taylor series. Recalling the fundamental commutator identity $[A, BC] = [A, B]C + B[A, C]$, we evaluate the commutator of a single annihilation operator with a pair creation operator:

$$[a_{\mathbf{k}}, a^\dagger_{\mathbf{p}} a^\dagger_{-\mathbf{p}}] = [a_{\mathbf{k}}, a^\dagger_{\mathbf{p}}]a^\dagger_{-\mathbf{p}} + a^\dagger_{\mathbf{p}}[a_{\mathbf{k}}, a^\dagger_{-\mathbf{p}}]$$
$$= a^\dagger_{-\mathbf{p}}\delta_{\mathbf{k},\mathbf{p}} + a^\dagger_{\mathbf{p}}\delta_{\mathbf{k},-\mathbf{p}}.$$

More generally, this can be easily verified by mathematical induction for arbitrary powers of the pair operator:

$$[a_{\mathbf{k}}, (a^\dagger_{\mathbf{p}})^n (a^\dagger_{-\mathbf{p}})^n] = [a_{\mathbf{k}}, (a^\dagger_{\mathbf{p}})^n](a^\dagger_{-\mathbf{p}})^n + (a^\dagger_{\mathbf{p}})^n[a_{\mathbf{k}}, (a^\dagger_{-\mathbf{p}})^n]$$
$$= n(a^\dagger_{\mathbf{p}})^{n-1}\delta_{\mathbf{k},\mathbf{p}}(a^\dagger_{-\mathbf{p}})^n + (a^\dagger_{\mathbf{p}})^n n(a^\dagger_{-\mathbf{p}})^{n-1}\delta_{\mathbf{k},-\mathbf{p}}$$
$$= \frac{\partial}{\partial a^\dagger_{\mathbf{k}}}\left((a^\dagger_{\mathbf{p}})^n (a^\dagger_{-\mathbf{p}})^n\right).$$

In this way, one can see that the operational action of the commutator is completely mathematically analogous to a formal differentiation with respect to the conjugate creation operator $a^\dagger_{\mathbf{k}}$. Leveraging this powerful result, we evaluate the commutator with the full exponential:

$$\left[a_{\mathbf{k}}, \exp\left(-\frac{1}{2}\sum_{\mathbf{p}}{}' \frac{v_p}{u_p} a^\dagger_{\mathbf{p}} a^\dagger_{-\mathbf{p}}\right)\right] = \frac{\partial}{\partial a^\dagger_{\mathbf{k}}} \exp\left(-\frac{1}{2}\sum_{\mathbf{p}}{}' \frac{v_p}{u_p} a^\dagger_{\mathbf{p}} a^\dagger_{-\mathbf{p}}\right)$$
$$= -\frac{1}{2}\sum_{\mathbf{p}}{}' \frac{v_p}{u_p}(\delta_{\mathbf{k},\mathbf{p}} a^\dagger_{-\mathbf{p}} + \delta_{\mathbf{k},-\mathbf{p}} a^\dagger_{\mathbf{p}}) \exp\left(-\frac{1}{2}\sum_{\mathbf{p}}{}' \frac{v_p}{u_p} a^\dagger_{\mathbf{p}} a^\dagger_{-\mathbf{p}}\right)$$
$$= -\frac{v_k}{u_k} a^\dagger_{-\mathbf{k}} \exp\left(-\frac{1}{2}\sum_{\mathbf{p}}{}' \frac{v_p}{u_p} a^\dagger_{\mathbf{p}} a^\dagger_{-\mathbf{p}}\right).$$

By substituting this evaluated commutator back into our initial expression, we obtain the exact relation:

$$u_k a_{\mathbf{k}} \exp\left(-\frac{1}{2}\sum_{\mathbf{p}}{}' \frac{v_p}{u_p} a^\dagger_{\mathbf{p}} a^\dagger_{-\mathbf{p}}\right)|0\rangle = -v_k a^\dagger_{-\mathbf{k}} \exp\left(-\frac{1}{2}\sum_{\mathbf{p}}{}' \frac{v_p}{u_p} a^\dagger_{\mathbf{p}} a^\dagger_{-\mathbf{p}}\right)|0\rangle,$$

which rigorously proves that our proposed exponential pair-state possesses exactly the property demanded by Equation 1.3.3. Note that the exponential operator acting on the vacuum creates an infinite superposition of bosonic pairs with equal and opposite non-zero momenta, serving as the formal quantum mechanical description of the condensate depletion.

Finally, let us address the macroscopic component corresponding to $\mathbf{k} = 0$. Recalling the mean-field substitution utilized to diagonalize the Hamiltonian ($a_0 \to \sqrt{N_0}$ and $a_0^\dagger \to \sqrt{N_0}$), we physically require that the expectation values of the macroscopic creation and annihilation operators over the true ground state equal this condensed value:

$$\langle \psi_0 | a_0 | \psi_0 \rangle = \sqrt{N_0},$$

$$\langle \psi_0 | a_0^\dagger | \psi_0 \rangle = \sqrt{N_0}.$$

These two expectation value equations are simultaneously satisfied by enforcing the eigenvalue relation:

$$a_0 | \psi_0 \rangle = \sqrt{N_0} | \psi_0 \rangle.$$

This defining equation is universally satisfied by Glauber coherent states, which are mathematically structured as:

$$e^{-N_0/2} e^{\sqrt{N_0} a_0^\dagger} | 0 \rangle.$$

Because the zero-momentum operators completely commute with the pair-creation operators at finite momenta ($\mathbf{p} \neq 0$), both the mean-field condensate requirement and the Bogoliubov pair-correlation requirement can be satisfied simultaneously. Collecting these distinct physical contributions, we arrive at the complete, unified expression for the interacting many-body ground state:

$$| \psi_0 \rangle = \frac{1}{\mathcal{N}} e^{\sqrt{N_0} a_0^\dagger} \exp \left(-\frac{1}{2} {\sum_{\mathbf{p}}}' \frac{v_p}{u_p} a_{\mathbf{p}}^\dagger a_{-\mathbf{p}}^\dagger \right) | 0 \rangle, \tag{1.3.4}$$

where $\mathcal{N}$ is an overall normalization constant that ensures $\langle \psi_0 | \psi_0 \rangle = 1$. By explicitly computing the norm of this state, this constant can be analytically proven to be:

$$\mathcal{N} = e^{N_0/2} \left(\prod_{\mathbf{k}} u_k^2 \right)^{1/2}.$$

1.4 | Spontaneous Symmetry Breaking and Goldstone Theorem

It is straightforward to verify that the original unperturbed Hamiltonian commutes with the total number operator $\hat{N} = \sum_{\mathbf{k}} a^{\dagger}_{\mathbf{k}} a_{\mathbf{k}}$. This commutation property is intimately connected with the fundamental invariance of the Hamiltonian under global U(1) phase transformations:

$$a_{\mathbf{k}} \to e^{i\phi} a_{\mathbf{k}},$$

$$a^{\dagger}_{\mathbf{k}} \to e^{-i\phi} a^{\dagger}_{\mathbf{k}}.$$

These phase rotations are formally realized by the application of the unitary continuous symmetry operator $U(\phi)$:

$$U(\phi) = e^{i\phi \hat{N}}.$$

Indeed, it is easy to explicitly check the generator action on the annihilation operator:

$$U^{\dagger}(\phi) a_{\mathbf{k}} U(\phi) = e^{i\phi} a_{\mathbf{k}}.$$

Because the Hamiltonian is invariant under this specific transformation—meaning $U^{\dagger}(\phi) H U(\phi) = H$—Noether's theorem guarantees there should be a conserved macroscopic charge. This conserved charge is nothing else than the mathematical generator of the transformation itself, namely, the total number operator $\hat{N}$.

Crucially, even if the Hamiltonian governing the system is perfectly invariant under the above U(1) transformation, the interacting ground state $|\psi_0\rangle$ is not. This profound physical phenomenon is known as spontaneous symmetry breaking. There are multiple analytical ways to see this symmetry breaking. Consider the macroscopic expectation value of the zero-momentum annihilation operator a_0. We previously established that over the coherent ground state, $\langle \psi_0 | a_0 | \psi_0 \rangle \neq 0$. Applying the unitary transformation, we have:

$$\langle \psi_0 | (U^{\dagger}(\phi) a_0 U(\phi)) | \psi_0 \rangle = \langle \psi_0 | (e^{i\phi} a_0) | \psi_0 \rangle.$$

However, if the ground state $|\psi_0\rangle$ were truly invariant under the symmetry group, then by definition $U(\phi)|\psi_0\rangle = |\psi_0\rangle$. This invariant assumption would require that the equation evaluates to:

$$\langle \psi_0 | a_0 | \psi_0 \rangle = e^{i\phi} \langle \psi_0 | a_0 | \psi_0 \rangle,$$

which can logically be satisfied for an arbitrary phase ϕ if and only if $\langle \psi_0 | a_0 | \psi_0 \rangle = 0$. This immediately leads to a direct contradiction with our foundational result that the condensate is macroscopically occupied. The exact same mathematical proof can be repeated for the pair correlation expectation value at finite momenta $\mathbf{k} \neq 0$. In that case, we notice that

$$\langle \psi_0 | a_{\mathbf{k}} a_{-\mathbf{k}} | \psi_0 \rangle = -u_k v_k \neq 0,$$

and applying the operator $U(\phi)$ to both bare operators and falsely imposing the invariance of the ground state $|\psi_0\rangle$ once again yields a fatal contradiction.

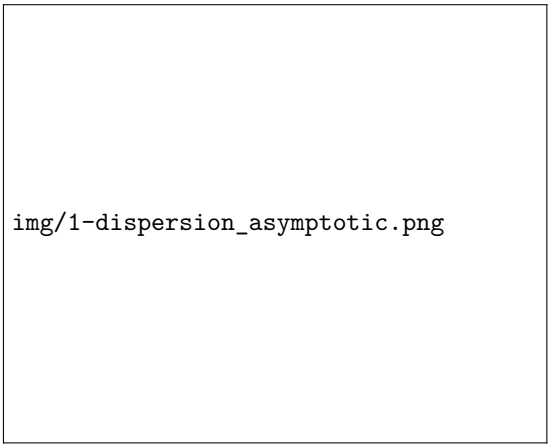
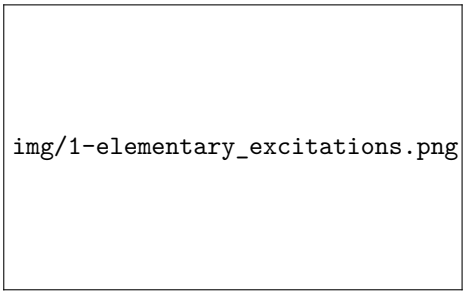


Figure 1.5: Plot of the dispersion $E_k = \sqrt{\epsilon_k^2 + 2nU_0\epsilon_k}$ (blue line) with its two asymptotic behaviours: for $k \to 0$ we have that the dispersion is linear with $E_k \sim c\hbar k$, while for $k \to \infty$ we have that the dispersion is quadratic with $E_k \sim \epsilon_k + nU_0$.

Once we have rigorously verified that the chosen ground state explicitly breaks the continuous U(1) symmetry, we know by Goldstone's theorem that there must emerge massless Goldstone bosons. These excitations must exhibit an energy dispersion that vanishes continuously (specifically, linearly) with the momentum p, yielding $E \sim cp$ for $p \to 0$. Recall that after executing the Bogoliubov transformation, the interacting Hamiltonian becomes diagonalized, featuring the quasiparticle energy spectrum given by E_k. The asymptotic limit $k \to 0$ for E_k has been already evaluated, successfully reproducing exactly the linear phononic dispersion that Goldstone's theorem demands: $E_k \sim c\hbar k$ (as visualized in Figure 1.5), with the characteristic sound velocity c scaling with the interaction strength.

Figure 1.6: Elementary excitations are modes that allow us to describe low energy states. If we excite the ground state GS, increasing the energy by $+E_k$, it's like creating an elementary excitation (a quasi-particle).



The derived energy values E_k structurally describe the dispersion of these elementary excitations, which are collective modes that allow us to describe effectively low-energy many-body states as a simple ideal gas of quasi-particles (see Figure 1.6). This quasiparticle framework is particularly relevant at very low temperatures T, when only these low-energy excited states are thermally activated, since random thermal fluctuations are small and strictly of order $k_B T$.

Consequently, complex interacting states at low energy can be cleanly described as a dilute gas of elementary excitations. It is critical to recognize that these independent excitations are not the original bare bosons created by the standard operators $a^\dagger_{\mathbf{k}}$, but rather new hybridized bosonic modes created by the Bogoliubov operators $\alpha^\dagger_{\mathbf{k}}$ for small k. These elementary excitations are governed by an effective interacting dispersion E_k that is structurally distinct from the purely quadratic free-particle dispersion ϵ_k of the original bare bosons.

Remarkably, even for a strongly interacting bare Bose gas, provided the temperature remains sufficiently low, very few of these hybridized modes are activated. Because the thermal density of quasi-particles is low, these elementary excitations are highly dilute and can thus be treated to an excellent approximation as a non-interacting ideal gas. Lastly, we note a fundamental statistical

constraint: for a system comprised exclusively of underlying bosons, like in the present model, only bosonic elementary excitations are physically possible. In contrast, for a many-body system of underlying fermions, the collective excitations can manifest as either fermionic (quasiparticles) or bosonic (e.g., Cooper pairs, plasmons) elementary excitations depending on the specific pairing mechanics.

1.5 | Specific Heat and Condensate Fraction

Let us proceed with the calculation of the specific heat of a dilute Bose gas. From the diagonalized Hamiltonian, we established that at absolute zero temperature, $T = 0$, the system resides in the quasiparticle vacuum with no elementary excitations. At a finite temperature $T > 0$, the thermal mean number of elementary excitations, $\langle \alpha_{\mathbf{k}}^{\dagger} \alpha_{\mathbf{k}} \rangle$, is governed by the standard Bose-Einstein distribution—since these Bogoliubov quasiparticles are fundamentally bosons:

$$\langle \alpha_{\mathbf{k}}^{\dagger} \alpha_{\mathbf{k}} \rangle = \frac{1}{e^{\beta E_k} - 1},$$

where $\beta = (k_B T)^{-1}$. In this expression, the chemical potential μ associated with the elementary excitations is strictly zero. This physically arises because the total number of quasiparticle excitations is not a conserved quantity; it is determined dynamically by the thermal equilibrium at temperature T (entirely analogous to phonons in a crystalline solid or photons in black-body radiation). Consequently, the total internal energy $E(T)$ of the gas at temperature T is given by the sum of the zero-point ground state energy and the thermal excitation energy:

$$E(T) = \mathcal{E}_0 + \sum_{\mathbf{k}} \frac{E_k}{e^{\beta E_k} - 1}.$$

Taking the thermodynamic limit, the discrete summation over wave vectors transitions into a continuous integral, yielding the total energy density:

$$\frac{E(T)}{V} = \frac{\mathcal{E}_0}{V} + \int \frac{d^3k}{(2\pi)^3} \frac{E_k}{e^{\beta E_k} - 1}.$$

For high-energy modes where $E_k \gtrsim k_B T$, the integrand is exponentially suppressed by the Bose distribution. Therefore, the dominant contribution to the integral comes strictly from the low-energy domain where $E_k \lesssim k_B T$. At very low temperatures, this restricts our focus to the extreme long-wavelength limit of the dispersion relation.

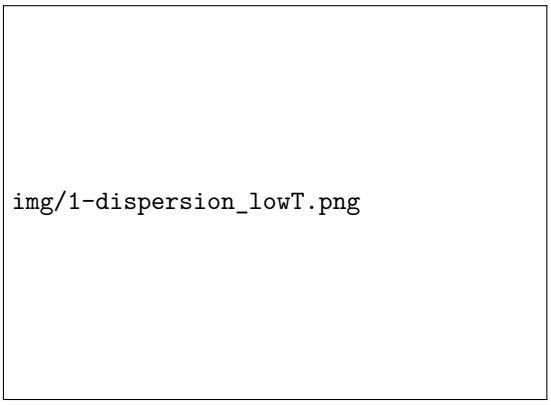


Figure 1.7: At low temperature T, if we are interested just in the region $E_k \lesssim k_B T$, we can see that the dispersion E_k goes linearly with the momentum k.

In this low-momentum regime, we recall that the Bogoliubov dispersion is strictly linear, $E_k \approx c\hbar k$. Thus, in the asymptotic limit $T \to 0$, we can safely replace the full dispersion relation with its phononic approximation inside the integral:

$$\frac{E(T)}{V} = \frac{\mathcal{E}_0}{V} + \int \frac{d^3k}{(2\pi)^3} \frac{c\hbar k}{e^{\beta c\hbar k} - 1}.$$

Evaluating this integral by switching to spherical coordinates and introducing the dimensionless integration variable $x = \beta c\hbar k$, we obtain:

$$\begin{aligned}\frac{E(T)}{V} &= \frac{\mathcal{E}_0}{V} + \left(\frac{k_B T}{c\hbar}\right)^3 k_B T \frac{4\pi}{(2\pi)^3} \int_0^\infty \frac{x^3}{e^x - 1} dx \\ &= \frac{\mathcal{E}_0}{V} + \frac{\pi^2}{30} \frac{(k_B T)^4}{(c\hbar)^3}.\end{aligned}$$

From this energy density, we directly compute the isochoric heat capacity, C_V:

$$C_V = \left(\frac{\partial E(T)}{\partial T}\right)_V = \frac{2\pi^2}{15} V k_B \left(\frac{k_B T}{c\hbar}\right)^3.$$



Figure 1.8: Plot of the behavior of the specific heat C_V at low temperature for a bosonic system. The blue line represents the specific heat of an interacting Bose gas, which scales proportionally to the cube of the temperature. The red line represents the specific heat of a non-interacting Bose gas, scaling proportionally to the temperature to the power of three halves.

This final expression reveals that the low-temperature specific heat of the weakly interacting Bose gas scales as T^3. This characteristic cubic temperature dependence mirrors the behavior of acoustic phonons in a Debye solid or photons in a cavity, fundamentally stemming from the linear phononic dispersion relation. We can starkly contrast this interacting result with the ideal, non-interacting Bose gas, which exhibits a $T^{3/2}$ dependence due to its purely quadratic free-particle dispersion. In the interacting system, the linear dispersion creates a lower density of states at low energies compared to the non-interacting case, making the elementary excitations significantly harder to thermally excite at temperatures near absolute zero.

1.6 | Landau Criterion for Superfluidity

In a normal fluid flowing through a pipe, friction between the fluid and the pipe boundaries leads to a continuous dissipation of energy and a corresponding macroscopic pressure drop along the direction of flow. Conversely, superfluidity (a remarkable quantum phenomenon famously discovered in liquid helium by Pyotr Kapitza) is characterized by the ability of certain fluids to flow without any friction. In a perfectly superfluid state, even at a finite macroscopic flow velocity $\mathbf{v} > 0$, the pressure remains exactly constant along the fluid, such that $\nabla p = 0$.

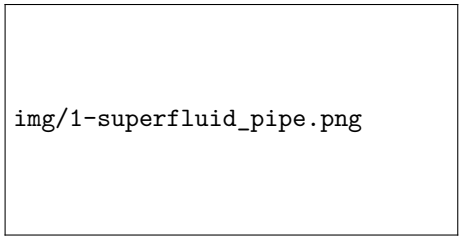


Figure 1.9: A fluid flows in a pipe at a velocity $\mathbf{v}$ and along the flow the pressure decreases due to viscosity, generating a gradient of pressure ∇p.

Our primary goal is to leverage the spectral properties of the many-body Hamiltonian and the associated elementary excitation dispersion relation to rigorously derive a fundamental condition known as the Landau criterion for superfluidity.

To understand the microscopic origin of friction, consider a classical many-body fluid, such as water, flowing through a pipe. If we assume the system is initially prepared in a state devoid of internal excitations, physical friction with the pipe boundaries will eventually induce their formation. Consequently, the total internal energy of the macroscopic system increases by the cumulative energy of these newly generated thermal excitations. For instance, if the initial state follows a classical Maxwell-Boltzmann momentum distribution, the dissipation of macroscopic forward momentum into random thermal excitations will eventually cause the distribution to narrow and shift toward lower net momenta. This physical picture implies that by examining the energy spectrum and the quasiparticle dispersion relation, we can determine precisely when adding energy to the system through such thermal or frictional processes is kinematically allowed.

1.6.1 | Kinematic Framework and Galilean Transformations

Lev Landau proposed an elegant kinematic criterion to establish whether a zero-temperature ($T = 0$) system behaves as a superfluid, specifically allowing one to predict the critical velocity at which the frictionless superfluid flow breaks down. Landau's derivation relies on comparing the energetics of the system in two distinct inertial reference frames: the laboratory frame K, where the fluid moves with a macroscopic velocity $\mathbf{v}$ and the pipe is perfectly at rest, and the co-moving frame K', where the fluid is strictly at rest and the pipe moves backwards with a velocity $-\mathbf{v}$. The frame K' thus translates with a velocity $\mathbf{v}$ with respect to the laboratory frame K.

In the co-moving frame K', the fluid is statically at rest. If the system experiences absolutely no friction with the pipe walls, it remains in its pristine many-body ground state, characterized by the complete absence of thermal excitations ($\langle \alpha^\dagger_{\mathbf{k}} \alpha_{\mathbf{k}} \rangle = 0$ for all $\mathbf{k}$). In this unperturbed scenario, the total energy of the fluid is simply its zero-point ground state energy, $E'_{(0)} = \mathcal{E}_0$.

If, however, friction were to occur, the interaction with the moving pipe walls would perturb the fluid, doing work to create elementary excitations. The system's internal energy would consequently

increase by $\sum_{\mathbf{k}} E_k \langle \alpha_{\mathbf{k}}^\dagger \alpha_{\mathbf{k}} \rangle$, where the expectation values $\langle \alpha_{\mathbf{k}}^\dagger \alpha_{\mathbf{k}} \rangle$ are positive integers (or strictly non-zero for finite-temperature statistical ensembles). To analyze the threshold of this dissipative process, let us consider the creation of a single elementary excitation with momentum $\mathbf{p}$. The total energy of the perturbed fluid in frame K' then becomes $E'_{(\mathrm{exc})} = \mathcal{E}_0 + E_p$.

We now transform this energy back to the laboratory frame K using the standard **Galilean transformation** for the total energy of a many-particle system:

$$E = E' + \frac{MV^2}{2} + \mathbf{P}' \cdot \mathbf{V},$$

where E is the energy in frame K, E' and $\mathbf{P}'$ are the total internal energy and total internal momentum in frame K', M is the total macroscopic mass of the fluid, and $\mathbf{V}$ is the relative velocity of K' with respect to K. In our setup, this relative frame velocity is exactly $\mathbf{V} = \mathbf{v}$.

Remark. *When performing this Galilean boost, one must be exceedingly careful regarding the definition of momentum if the system is coupled to an external gauge field (such as an electromagnetic field for charged superfluids like superconductors). In such cases, the canonical momentum is shifted by the vector potential and does not simply correspond to the inertial mass times the velocity. For our neutral bosonic superfluid, however, this complication does not arise.*

For the fluid hosting a single elementary excitation, its internal energy in K' is $E' = E'_{(\mathrm{exc})} = \mathcal{E}_0 + E_p$. Because the macroscopic fluid is at rest in K', the only contribution to the total internal momentum comes from the newly created excitation itself, so $\mathbf{P}' = \mathbf{p}$. Substituting these into the transformation yields the total energy of the excited fluid in the laboratory frame K:

$$E_{(\mathrm{exc})} = \mathcal{E}_0 + E_p + \frac{Mv^2}{2} + \mathbf{p} \cdot \mathbf{v}.$$

Conversely, if the fluid remains in the unperturbed ground state, its internal energy in K' is merely $E' = E'_{(0)} = \mathcal{E}_0$, and its total internal momentum is identically zero ($\mathbf{P}' = 0$). Substituting these into the Galilean transformation, the energy of the frictionless flowing fluid in frame K is simply:

$$E_{(0)} = \mathcal{E}_0 + \frac{Mv^2}{2}.$$

Combining the last two expressions, we determine the energy cost to create the elementary excitation in the laboratory frame:

$$E_{(\mathrm{exc})} - E_{(0)} = E_p + \mathbf{p} \cdot \mathbf{v}.$$

To definitively establish whether the system behaves as a superfluid, we must examine the sign of this energy difference. Friction fundamentally occurs when the creation of an excitation is energetically favorable, which happens when the perturbed state has a lower total energy than the unperturbed flow ($E_{(\mathrm{exc})} < E_{(0)}$). In contrast, robust superfluidity is maintained whenever the spontaneous creation of excitations is kinematically forbidden because it costs energy ($E_{(\mathrm{exc})} > E_{(0)}$).

The critical velocity v_c is defined as the minimum macroscopic flow velocity at which the system's total energy can be spontaneously lowered (or kept equal) by the addition of elementary excitations. This threshold dictates exactly when it becomes thermodynamically convenient to dissipate kinetic energy. By imposing the boundary equality $E_{(\mathrm{exc})} = E_{(0)}$, we will obtain the precise conditions for the critical velocity v_c, which sharply separates the dissipative friction regime from the stable frictionless regime, thereby marking a dynamical phase transition to superfluidity.

1.6.2 | The Critical Velocity Threshold

In order for the spontaneous creation of an excitation to be kinematically possible in the laboratory reference frame K, the energy difference $E_{(\text{exc})} - E_{(0)}$ must be negative. This is because friction physically transfers kinetic energy from the moving fluid to the stationary pipe; consequently, the macroscopic fluid must decrease its total energy in this dissipative process. Friction will thus occur only if, for some values of the excitation momentum $\mathbf{p}$, the following inequality holds:

$$E_p + \mathbf{p} \cdot \mathbf{v} < 0.$$

The above condition is most easily satisfied if the created momentum $\mathbf{p}$ is aligned exactly anti-parallel (opposite) to the flow velocity $\mathbf{v}$. In this optimal dissipative configuration, the vector dot product becomes $-pv$, resulting in the simplified scalar condition:

$$\frac{E_p}{p} < v.$$

So, for a given macroscopic flow velocity v, friction will be physically possible if and only if the above condition is satisfied by some value of the excitation momentum p.

To find the absolute threshold for dissipation, we must consider the global minimum value of the ratio E_p/p as the momentum p is varied over all possible accessible states:

$$v_c \equiv \min_p \frac{E_p}{p}. \tag{1.6.1}$$

According to our kinematic inequality, only if the velocity v of the fluid strictly exceeds this critical threshold v_c is the creation of excitations (and thus friction) possible. We therefore conclude the fundamental Landau criterion: if v_c is strictly greater than zero, then the fluid will flow without any friction for all velocities $v < v_c$, behaving as a true superfluid. Mathematically, at a local minimum of the ratio E_p/p, simple calculus dictates that its derivative must vanish:

$$\frac{d}{dp}\left(\frac{E_p}{p}\right) = 0,$$

yielding the geometric condition:

$$\frac{dE_p}{dp} = \frac{E_p}{p}.$$

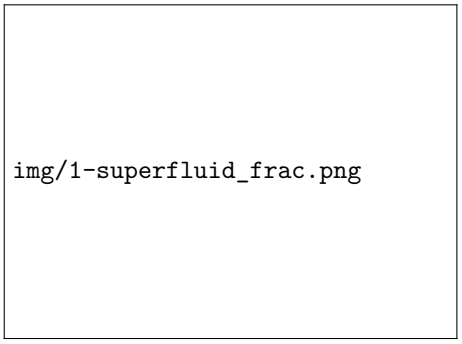


Figure 1.10: Plot of the dispersion E_p as a function of the momentum p (the blue line). For a given generic point P of coordinates (p, E_p), the two red lines represent respectively the tangent to the curve in P (with slope $\frac{dE_p}{dp}$) and the straight line connecting the origin to P, with slope E_p/p.

Note that this derivative equation has a highly intuitive geometric interpretation: if one plots the excitation energy E_p as a function of the momentum p, the derivative dE_p/dp represents the local slope of the tangent to the curve at the point (p, E_p), while the ratio E_p/p represents the overall

slope of the secant line joining the origin directly to that same point (p, E_p) (see Figure 1.10). When these two distinct slopes perfectly coincide, the secant line from the origin becomes tangent to the dispersion curve. It is exactly at this point of tangency that a minimum of E_p/p occurs, and the numerical value of this tangent slope identically defines the critical velocity v_c.

1.6.3 | Application to Physical Bose Systems

It is now time to apply the Landau criterion to various physical Bose systems, including both ideal and weakly interacting dilute gases, as well as strongly interacting systems like liquid helium. By analyzing the dispersion relations of elementary excitations in these systems, we can determine their critical velocities and assess their superfluid properties.

Ideal and Weakly Interacting Dilute Gases

The ideal Bose gas is a particularly instructive example. In this case, the energy dispersion is purely kinetic, given by $E_p = \frac{p^2}{2m}$. Let us analyze this system through the following example.

Example (No Superfluidity for Ideal BEC). Let us apply the Landau criterion to an ideal, non-interacting Bose-Einstein condensate. For a free particle, the energy dispersion is purely kinetic, meaning $E_p = \frac{p^2}{2m} \sim p^2$. When evaluating the critical velocity ratio, we find $\frac{E_p}{p} \propto \frac{p^2}{p} = p$. Taking the limit to find the minimum over all accessible momenta, $\min_p \left(\frac{E_p}{p}\right) \xrightarrow{p\to 0} 0$. Because $v_c = 0$, any arbitrarily small flow velocity $v > 0$ satisfies the condition for dissipation. It is therefore always kinematically convenient to create excitations, meaning an ideal BEC strictly exhibits no superfluidity.

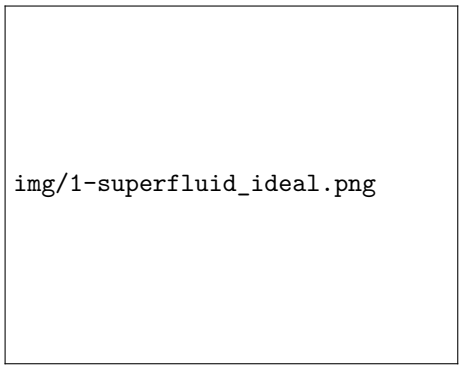


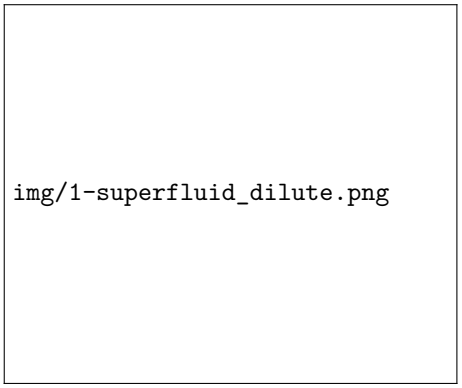
Figure 1.11: Dispersion E_p of elementary excitations in an ideal Bose gas as a function of the momentum p (blue line). The point in which the Landau criterion is satisfied is exactly at the origin, and the slope of the tangent in this point (horizontal red line) gives the critical velocity $v_c = 0$.

For the weakly interacting dilute Bose gas analyzed previously via Bogoliubov theory, recalling the behavior of E_p for small p, the dispersion is linear: $E_p \sim cp$. Consequently, the ratio E_p/p is a constant c near the origin. The global minimum condition is realized precisely as $p \to 0$ (see Figure 1.12):

$$v_c = \min_p \frac{E_p}{p} = c.$$

This profound result means that the weakly interacting Bose gas is a true superfluid for any flow velocity $v < c$. Only for supersonic flows $v > c$ are Bogoliubov quasiparticle excitations created, and macroscopic dissipation becomes possible.

Figure 1.12: Dispersion E_p of elementary excitations in a dilute Bose gas as a function of the momentum p (blue line). The point in which the Landau criterion is satisfied is the origin, and the slope of the tangent in this point (red line) gives the critical velocity: $v_c = \min_p E_p/p$. For $v \geq v_c$ there is friction, while for $v < v_c$ the gas is superfluid.



Superfluidity is not a universal guarantee; there will always be some critical parameter (such as a threshold velocity or a critical temperature) beyond which the system will spontaneously start to generate excitations and subsequently lose its superfluid properties.

Strongly Correlated Regimes: Liquid Helium-4 and Rotons

In strongly interacting Bose systems, such as liquid Helium-4, the situation becomes more complex. The measured spectrum of elementary excitations in liquid ^{4}He reveals a non-trivial dispersion relation that includes a local minimum at finite momentum, known as the "roton minimum". This feature significantly influences the critical velocity for superfluidity.

Example (Helium-4). Even in strongly interacting Bose systems that defy simple perturbative treatment, like liquid Helium-4, robust superfluid behavior emerges at sufficiently low temperatures. Finally, let us consider the measured spectrum of elementary excitations for liquid ^{4}He (as determined directly from neutron scattering experiments). We see that in this complex strongly-correlated case, there are actually two distinct geometric solutions (points of tangency) satisfying the Landau condition.

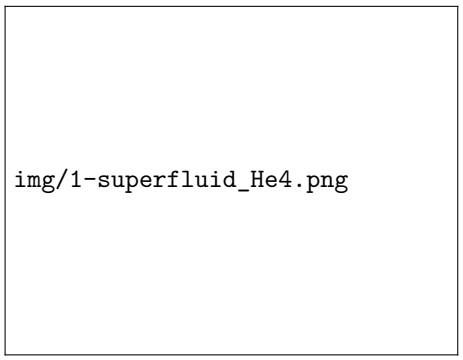


Figure 1.13: Dispersion E_p of elementary excitations in ^{4}He as a function of the momentum p (blue line). The points in which the Landau criterion is satisfied are two (straight red lines tangent to the curve). The one with the lowest slope determines the true critical velocity $v_c \approx 60$ m/s (while the initial sound velocity is much higher at $c = 240$ m/s). The gas is a superfluid for $v < v_c$.

The tangent line with the absolute lowest slope geometrically corresponds to the true global minimum of E_p/p. In Helium-4, this absolute minimum is reached not at the origin, but at a finite momentum close to the famous "roton minimum" of the dispersion curve E_p. The slope of this tangent gives the corresponding critical velocity for Helium-4: $v_c \approx 60$ m/s. Notice that this critical threshold for creating rotons is significantly lower than the speed of sound determined by the initial linear slope at the origin, which is $c = 240$ m/s.

The energy gap at this local minimum is called the roton gap, denoted Δ, and is experimentally

measured such that $\Delta/k_B = 8.7$ K. The elementary excitations grouped close to this finite-momentum minimum acquire a distinctly quadratic, particle-like effective form:

$$E_p = \Delta + \frac{(p - p_0)^2}{2m^*}$$

with an effective mass $m^* = 0.16m$ (where m is the bare mass of a ^{4}He atom) and a characteristic momentum $p_0/\hbar = 1.9$ Å^{-1}. These specific high-momentum elementary excitations are called rotons (while the low-momentum excitations corresponding to the initial linear part of the spectrum are standard acoustic phonons). We thus see that when the macroscopic superfluid velocity v exceeds $v_c \approx 60$ m/s, the spontaneous kinematic excitation of rotons completely destroys the superfluid behavior, even though the flow remains highly subsonic relative to c.

1.6.4 | The Macroscopic Probe and the Superfluid Phase Boundary

To deeply understand the physical mechanism underpinning the Landau argument, it is essential to clarify the asymmetric roles of the macroscopic boundary (e.g. the pipe or a moving capillary) and the quantum fluid. *Friction* in this context arises exclusively when the macroscopic motion of the object relative to the fluid creates elementary quasiparticle excitations within the fluid. At absolute zero temperature ($T = 0$), the superfluid resides entirely in its many-body ground state. Consequently, it cannot dissipate energy by transitioning to a lower energy state: it can only act as an **energy sink**, absorbing energy from the moving object by becoming excited.

In this framework, the pipe is treated strictly as an inert, classical macroscopic body moving at a well-defined velocity **v**. Its internal quantum spectrum, structure, or thermal kinetic energy are completely irrelevant to the onset of the dispersion process. The generation of friction is strictly a consequence of the fluid's capacity to generate excitations. As dictated by the Landau criterion, this unidirectional transfer of kinetic energy from the moving macroscopic probe to the quantum fluid becomes kinematically permitted only when the object's velocity exceeds the critical threshold v_c.

This critical boundary can be visualized geometrically in a parameter space plotting the temperature T against the flow velocity parameter v. The phase boundary cleanly separates the parameter space into a normal dissipative regime and a protected superfluid phase.

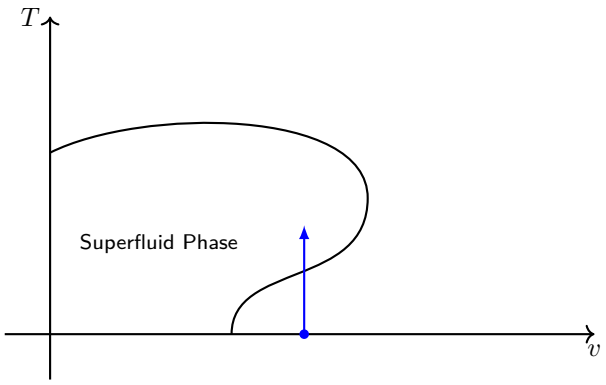


Figure 1.14: Schematic phase diagram illustrating the superfluid boundary as a function of the flow velocity parameter v and the temperature T. The solid curve defines the onset of excitation creation (friction). The blue arrow depicts a specific constant-velocity trajectory traversing the parameter space.

1.6.5 | Finite Temperature Effects and the Two-Fluid Model

It is crucial to note that the entire Landau derivation presented above assumes absolute zero temperature. If we consider a finite temperature $T > 0$, we must take into account the pre-existing gas of thermal excitations that are statistically present in the system. Indeed, even if the kinematic line pv sits safely below the zero-temperature critical dispersion threshold, we must consider the energy contributions and collisions from these thermal excitations, which can effectively transfer momentum and turn off the macroscopic superfluidity by traversing the dispersion gap. Consequently, the true critical velocity will be strictly lower than at zero temperature, and it will exhibit a strong dependence on the ambient temperature of the system.

Furthermore, at finite temperatures, there will always be a "normal" fraction of the fluid comprised of these thermal excitations that experiences ordinary friction. However, there concurrently exists a remaining fraction of the fluid that remains perfectly superfluid. According to the two-fluid model, the fraction of the fluid that is superfluid scales directly with the total number of particles, ensuring it remains a macroscopically sizable and observable portion of the system up until the critical transition temperature is reached.

Concluding Remarks: The Physics of Emergence

Through the mathematical machinery of the Bogoliubov transformation, we systematically mapped the bare bosonic degrees of freedom, $a_{\mathbf{k}}$, to a new set of operators, $\alpha_{\mathbf{k}}$. This was not merely a convenient mathematical change of basis, but a profound physical restructuring of the system. The new operators create and destroy collective, hybridized modes that intermix particles and holes. These emergent entities are defined as **quasiparticles**, and they—rather than the original bare bosons—constitute the true elementary excitations of our interacting many-body description.

This conceptual shift perfectly illustrates the celebrated paradigm articulated by P. W. Anderson: "*More is Different*". Quantum Many-Body Theory (QMBT) demonstrates that macroscopic physics is fundamentally distinct from single-particle quantum mechanics. Merely sticking individual constituent pieces together is entirely insufficient to understand the whole. As the system grows to the thermodynamic limit, non-trivial interactions generate entirely new emergent phenomena that simply do not exist at the level of isolated single particles.

A quintessential example of this macroscopic emergence is **spontaneous symmetry breaking**, which serves as a foundational pillar of modern QMBT. As we have seen, while the original Hamiltonian inherently respects a continuous global U(1) phase symmetry, the macroscopic interacting ground state spontaneously breaks it by developing a definite coherent phase. According to Goldstone's theorem, this broken continuous symmetry mathematically necessitates the emergence of massless Goldstone bosons. In the context of the interacting Bose gas, these Goldstone modes physically manifest as the acoustic phonons characterized by the linear dispersion relation, $E_k \sim c\hbar k$, at low momenta.

Ultimately, by closely analyzing the experimental observables and the low-energy spectrum of this system, we arrive at a powerful concluding lesson: **weak interactions can do a lot**. It is tempting to assume that exotic phenomena require strong, non-perturbative couplings. However, we have demonstrated that even an arbitrarily weak interaction is sufficient to completely rewire the vacuum state of the system. A profoundly rich phenomenology (encompassing quantum condensate depletion at absolute zero, the formation of a well-defined sound velocity, spontaneous symmetry breaking, and the onset of macroscopic superfluidity governed by the Landau criterion) emerges not from strong forces, but from the collective many-body nature of weakly interacting quasiparticles.

2 | BCS Theory of Superconductivity

Superconductivity was experimentally observed well before the formal discovery of quantum mechanics, presenting a profound physical mystery that eventually necessitated the development of a comprehensive quantum many-body description. The definitive microscopic explanation of this phenomenon, known as the BCS theory, was developed in 1957 by John Bardeen, Leon Cooper, and John Robert Schrieffer. At its core, the theory explains the superconducting state through the formation of bound states of electrons, known as Cooper pairs.

To formally develop this, we will consider a general system composed of spin-1/2 fermions experiencing a mutual *attractive* interaction between particles of opposite spin. Our primary theoretical interest is to determine the low-temperature phase of this many-body system. The central idea is that this mutual attraction can induce the formation of Cooper pairs—pairs of fermions with opposite spin. Because these composite pairs possess an integer total spin, they behave effectively as bosons and can undergo macroscopic condensation at sufficiently low temperatures.

The original motivation for studying such an interacting fermionic system was the microscopic explanation of metallic superconductivity. Superconductivity is a physical phenomenon occurring in certain metals and compounds whereby an electric current flows with absolutely zero electrical resistance below a specific critical temperature T_c. This phenomenon is deeply analogous to superfluidity; whereas a superfluid involves the macroscopic flow of a fluid with zero viscosity (the mechanical analog of electrical resistance) below a critical temperature, a superconductor involves the frictionless flow of charged carriers, namely electrons.

A crucial thermodynamic clue pointing toward the existence of these bound pairs of fermions in a superconductor is provided by the electronic specific heat, c_v. For an ideal Fermi gas at low temperatures T—which serves as a highly effective model for the conduction electron gas in a normal solid—the specific heat scales linearly with T. However, when a material transitions into a superconductor, the specific heat is observed to vanish exponentially with decreasing temperature. This exponential suppression is the hallmark thermodynamic behavior expected when there is a finite energy gap in the excitation spectrum separating the macroscopic ground state from the first available excited states. The magnitude of this energy gap can be extracted directly from specific heat measurements, revealing it to be extremely small—typically several orders of magnitude smaller—compared to the Fermi energy E_F of the electrons. It is natural to guess that this energy gap is directly associated with the finite binding energy of the fermion pairs. (The microscopic mechanism by which normally repulsive electrons can experience an effective mutual attraction in a metal lattice will be discussed in detail in the subsequent section).

Beyond its foundational success in condensed matter systems, the BCS theoretical framework we are going to develop is remarkably universal. The collection of phenomena that can be governed by these same equations spans an astonishing range of more than ten orders of magnitude in temperature. At the lower extreme, in the nanokelvin regime, the theory applies directly to ultracold Fermi gases trapped in optical lattices. These are dilute, neutral systems of fermionic atoms where an effective mutual attraction—arising fundamentally from van der Waals forces—can be accurately modeled as a simple contact interaction due to the extreme diluteness of the gas. These highly tunable systems have been studied very actively over recent years because experimentalists can artificially tune the strength of the interparticle interaction at will. The theory is equally relevant to other macroscopic neutral systems, such as liquid ^{3}He, where fermionic neutral atoms also experience a mutual attraction. At the absolute opposite extreme of the temperature and density scale, the very same BCS formalism can be extended to describe the pairing of quarks inside the ultra-dense, extremely high-temperature cores of neutron stars (a phenomenon known as color superconductivity).

2.1 | Basic Model for Effective Attraction

Let us consider a simplified model of a solid consisting of a cubic lattice of positively charged ions with a lattice spacing a, permeated by a gas of itinerant electrons. We assume the system is isolated, without any external electric or magnetic fields. The electrons are subject to a periodic potential generated by the ionic lattice; consequently, the potential minima are located in the immediate vicinity of the positive ions, marking the regions where the electrons are most likely to be found.

The electrons interact not only with this periodic ionic potential but also with one another via the standard Coulomb repulsion, $V_{e-e} \sim e^2/r$. However, the presence of the mobile electron gas embedded in the positively charged background activates the many-body phenomenon of screening. In a metal, the spatial redistribution of conduction electrons around any single negative charge effectively screens that negative charge at a distance from it. This screening effect drastically modifies the long-range Coulomb repulsion into a short-range effective interaction, which decays exponentially as $\sim e^{-r/\xi}$, where the screening length ξ is typically on the order of a few lattice spacings a. Although this screened interaction is significantly weaker at macroscopic distances, it remains fundamentally repulsive. To explain the formation of Cooper pairs, we must identify a physical mechanism capable of overcoming this residual repulsion to produce a net effective attraction between the electrons.

The Electron-Phonon Interaction and Fröhlich Mechanism

This requisite attraction arises from the interaction between the electrons and the collective vibrations of the ionic lattice, known as phonons. Because the typical Debye temperature characterizing these macroscopic phonon modes is on the order of $\omega_D \sim 100\,\mathrm{K}$, while superconducting phenomena typically occur at cryogenic temperatures around $1\,\mathrm{K}$, the thermal phonons are effectively frozen out. Consequently, we can safely treat the quantized phonon field as a perturbation to the interacting electron system.

As famously demonstrated by H. Fröhlich, in certain metals, the combination of the screening of the Coulomb repulsion and the virtual exchange of phonons between electrons may lead to an overall effective attraction between them. Specifically, the dynamic interaction between electrons and phonons produces a retarded attractive force. When an electron moves through the crystal, it locally modifies the ionic lattice by pulling the positive ions slightly toward its trajectory, creating a localized cloud of positive charges, and then rapidly moves away. Subsequently, a second electron may be electrostatically attracted by this localized accumulation of positive charge.

A key physical ingredient of this retarded mechanism is the stark difference in dynamical time scales: the heavy distortions of the ionic lattice fade away rather slowly in comparison with the extremely fast time scale over which one electron is replaced by a second one at a certain position due to the rapid motion of the electrons. Ultimately, these two combined effects—electronic screening and virtual phonon exchange—may lead in certain materials to an overall effective attraction between the electrons, provided those electrons belong to a tiny energy shell situated immediately around the Fermi energy E_F.

2.1.1 | Effective Interaction

We aim to describe the behavior of electrons in a solid and understand how they interact with the lattice vibrations. A solid can be modeled as a static lattice of ions, with electrons occupying localized orbitals and tunneling through them. The classical positions of the ions are perturbed by dynamical excitations, which we describe as **phonons**.

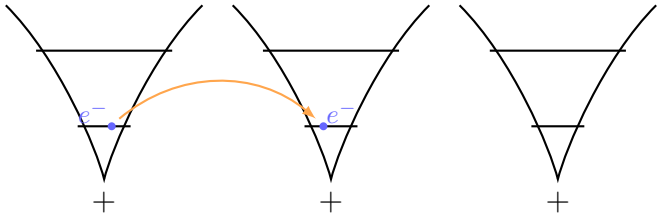


Figure 2.1: Schematic representation of a solid modeled as a static lattice of positive ions. The electrons occupy localized orbitals within the ionic Coulomb potential wells and can tunnel (hop) to adjacent sites. This baseline periodic structure is subsequently perturbed by dynamical lattice vibrations (phonons), which modulate the effective electron-electron interactions.

In the harmonic approximation, the phonon Hamiltonian is constructed as a sum over independent harmonic oscillators:

$$H_{\text{ion}} = \frac{1}{2}\sum_{\mathbf{q}} \hbar\omega_{\mathbf{q}} a^\dagger_{\mathbf{q}} a_{\mathbf{q}},$$

where $a^\dagger_{\mathbf{q}}$ and $a_{\mathbf{q}}$ are the bosonic creation and annihilation operators for a phonon with momentum $\mathbf{q}$, and $\omega_{\mathbf{q}}$ is the phonon frequency.

The Hamiltonian for the electrons is composed of an interaction part and a kinetic part:

$$H_{\text{el}} = H_{\text{int}} + H_{\text{kin}}.$$

Interaction Hamiltonian. Naively, the interaction between electrons is the bare Coulomb repulsion. However, the presence of a background of positive charges screens this interaction. The effective screened potential is often described by the Thomas-Fermi model, which introduces an exponential decay over the distance $|\mathbf{r}|$:

$$V(\mathbf{r}) \propto \frac{e^{-|\mathbf{r}|/\xi}}{|\mathbf{r}|} \sim e^{-|\mathbf{r}|/\xi},$$

where ξ is the Thomas-Fermi screening length.

While the kinetic energy can be formulated on a real-space lattice (yielding the Hubbard model), it is much more convenient to diagonalize it directly in momentum space. Assuming that the kinetic energy is independent of the electron spin σ, we have:

$$H_{\text{kin}} = \sum_{\mathbf{k},\sigma} \epsilon_{\mathbf{k}} c^\dagger_{\mathbf{k},\sigma} c_{\mathbf{k},\sigma},$$

where $c^\dagger_{\mathbf{k},\sigma}$ and $c_{\mathbf{k},\sigma}$ are the fermionic creation and annihilation operators for an electron with momentum $\mathbf{k}$ and spin σ, and $\epsilon_{\mathbf{k}} = \frac{\hbar^2 k^2}{2m}$ is the bare electron dispersion.

Total Hamiltonian. Combining the lattice and electronic components, the total Hamiltonian is:

$$H = H_{\text{ion}} + H_{\text{int}} + H_{\text{kin}}.$$

Our ultimate goal is to use perturbation theory to trace out the phonon degrees of freedom. This procedure will yield an effective Hamiltonian that only contains the fermionic operators $(c, c^\dagger)$, effectively incorporating the phonons as a mediated interaction between the electrons:

$$H \to H_{\text{eff}}(c, c^\dagger).$$

2.1.2 | Single Particle Problem

Electron-Phonon Interaction. How can an electron interact with a phonon? By traveling through the solid, an electron can kick the ions and excite their physical modes. Based on the principle of Occam's razor, we consider only the simplest fundamental processes: the emission and the absorption of a single phonon.

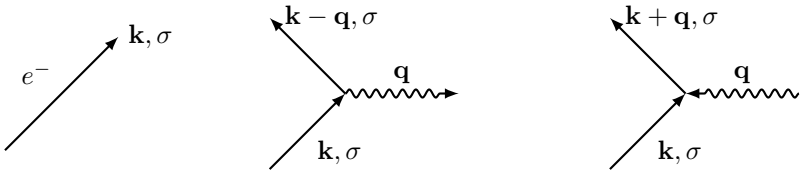


Figure 2.2: Fundamental electron-phonon interaction processes. Left: A free electron propagating with momentum $\mathbf{k}$ and spin σ. Middle: Emission of a single phonon with momentum $\mathbf{q}$, scattering the electron to a new momentum state $\mathbf{k}-\mathbf{q}$. Right: Absorption of a single phonon with momentum $\mathbf{q}$, scattering the electron to momentum $\mathbf{k}+\mathbf{q}$. In both interactions, the spin σ is conserved.

The phenomenological interaction potential V_1 describing these scattering processes must contain a sum over all possible electron momenta $\mathbf{k}$, spins σ, and phonon momenta $\mathbf{q}$:

$$V_1 = \sum_{\mathbf{k},\sigma} \sum_{\mathbf{q}} \left(M_{\mathbf{q},\sigma} a_{\mathbf{q}} c^\dagger_{\mathbf{k}+\mathbf{q},\sigma} c_{\mathbf{k},\sigma} + M_{-\mathbf{q},\sigma} a^\dagger_{\mathbf{q}} c^\dagger_{\mathbf{k}-\mathbf{q},\sigma} c_{\mathbf{k},\sigma} \right).$$

The amplitude of this interaction is governed by the matrix element $M_{\mathbf{q},\sigma}$, which takes the form:

$$M_{\mathbf{q},\sigma} \simeq \frac{|\mathbf{q}|}{\sqrt{M_{\text{ion}} \omega_{\mathbf{q}}}}.$$

Physically, $M_{\mathbf{q},\sigma}$ is inversely proportional to the square root of the ion mass M_{ion}. If the lattice is composed of heavy ions, it is extremely challenging for the electrons to excite the phononic modes, causing the matrix element to vanish rapidly. Conversely, lighter ions allow for an easier transfer of energy from the electrons to the ion chain.

Perturbation Theory in V_1. Our full system is described by:

$$H = H_{\text{kin}} + H_{\text{ph}} + V_1.$$

We proceed by treating V_1 as a perturbation. To do so, we must first define the unperturbed vacuum (ground state) for both fermions and bosons.

Unperturbed Vacuum. We evaluate the vacuum in the presence of some initial particle excitations. This state, $|\phi_{n_e,n_{ph}}\rangle$, is formed by the tensor product of the electron and phonon states:

$$|\phi_{n_e,n_{ph}}\rangle = \left(\prod_{\substack{\mathbf{k}_j=\mathbf{k}_1,\ldots,\mathbf{k}_{n_e} \\ \sigma_j=\sigma_1,\ldots,\sigma_{n_e}}} c^\dagger_{\mathbf{k}_j,\sigma_j} \right) \left(\prod_{\mathbf{q}_j=\mathbf{q}_1,\ldots,\mathbf{q}_{n_{ph}}} a^\dagger_{\mathbf{q}_j} \right) |0,0\rangle.$$

Here, the absolute vacuum is defined as $|0,0\rangle = |n_e = 0, n_{ph} = 0\rangle$. The phonon vacuum itself is a tensor product over all possible momentum states:

$$|0,0\rangle = |0\rangle_{\mathbf{q}_1} \otimes |0\rangle_{\mathbf{q}_2} \otimes \ldots$$

Because the system is confined in a box/lattice, the allowed momenta are discrete and defined within the Brillouin zone. For example, along the x-direction:

$$q_x^{(J)} = \frac{2\pi J}{L_x}, \quad J \in \mathbb{Z}.$$

For the electrons, this highly populated unperturbed ground state is commonly referred to as the **Fermi Sea**.

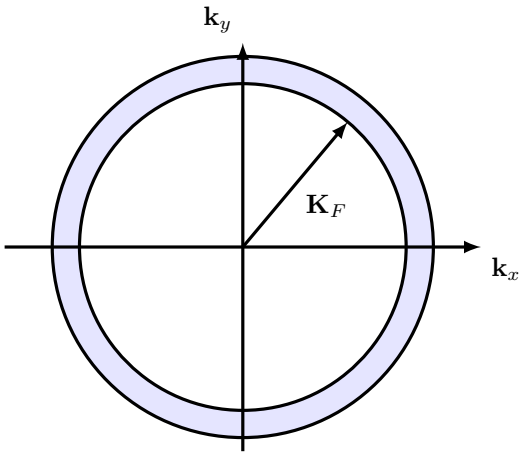


Figure 2.3: Two-dimensional representation of the momentum space for the unperturbed electron gas. The highly populated ground state forms a Fermi sea, bounded by the Fermi circle of radius K_F. The shaded narrow shell highlights the tiny energy window around the Fermi surface where the effective electron-electron interaction, mediated by phonons, is non-zero and attractive.

First Order Perturbation Theory

Let us compute the first-order correction to the energy, which fundamentally requires evaluating the expectation value of the perturbation V_1 over the unperturbed vacuum state $|\phi_0\rangle$:

$$E_1 = \langle\phi_0|V_1|\phi_0\rangle.$$

If we define our initial state explicitly by its definite number of electrons and phonons, $|\phi_0\rangle = |n_e, n_{ph}\rangle$, applying the interaction potential V_1 physically yields a new state where the total number of phonons has changed by exactly one (due to the fundamental processes of either a single emission or a single absorption):

$$V_1|\phi_0\rangle = |n_e, n_{ph} \pm 1\rangle.$$

Because the number of phonons strictly changes under the action of V_1, the resulting state is mathematically orthogonal to the initial state. Under the physically motivated assumption that the phonons do not macroscopically condense (which would give them a finite vacuum expectation value), the first-order energy correction strictly vanishes:

$$\implies \langle n_e, n_{ph}|n_e, n_{ph} \pm 1\rangle = 0.$$

Second Order Perturbation Theory

Since the leading first-order term is identically zero, we must evaluate the perturbation to second order to capture the leading interaction effects. The second-order energy correction E_2 is formally given by summing over all possible intermediate states $|\phi_{n,m}\rangle$:

$$E_2 = \sum_{n,m} \frac{\langle\phi_0|V_1|\phi_{n,m}\rangle\langle\phi_{n,m}|V_1|\phi_0\rangle}{E_0^{(\text{unp.})} - E_{n,m}^{(\text{unp.})}}.$$

Out of this seemingly infinite sum over all basis states, only highly specific terms survive the projection. The number of electrons n in the intermediate state must strictly remain equal to n_e, and the number of phonons m must be exactly $n_{ph}+1$ or $n_{ph}-1$. This selection rule allows us to simplify the numerator structure using Kronecker deltas:

$$E_2 = \sum_{n,m} \frac{\delta_{n,n_e}\left(\delta_{m,n_{ph}+1} + \delta_{m,n_{ph}-1}\right)}{E_0^{(\text{unp.})} - E_{n,m}^{(\text{unp.})}}.$$

Example (Single electron application). To intimately understand the underlying physics, let us perform the explicit calculation for the simplest possible case: a single electron with initial momentum $\mathbf{p}$ and spin σ, and exactly zero initial phonons ($n_e = 1$, $n_{ph} = 0$). The initial state is constructed as $|\phi_0\rangle = c^\dagger_{\mathbf{p},\sigma}|0\rangle$. Applying the interaction V_1 to this specific state and carefully enforcing both momentum and spin conservation through the resulting delta functions, we obtain:

$$\begin{aligned} V_1|\phi_0\rangle &= \sum_{\mathbf{q},\sigma'}\sum_{\mathbf{k}'} M_{\mathbf{q}}(a^\dagger_{\mathbf{q}} + a_{-\mathbf{q}})c^\dagger_{\mathbf{k}'+\mathbf{q},\sigma'}c_{\mathbf{k}',\sigma'}c^\dagger_{\mathbf{p},\sigma}|0\rangle \\ &= \sum_{\mathbf{q},\sigma'}\sum_{\mathbf{k}'} M_{\mathbf{q}}(a^\dagger_{\mathbf{q}} + a_{-\mathbf{q}})c^\dagger_{\mathbf{k}'+\mathbf{q},\sigma'}\delta_{\sigma,\sigma'}\delta_{\mathbf{k}',\mathbf{p}}|0\rangle \\ &= \sum_{\mathbf{q}} M_{\mathbf{q}}(a^\dagger_{\mathbf{q}} + a_{-\mathbf{q}})c^\dagger_{\mathbf{p}+\mathbf{q},\sigma}|0\rangle. \end{aligned}$$

This rigorous evaluation explicitly shows the initial electron's momentum being shifted by $\mathbf{q}$ due to the interaction with the phonon field.

Now, we compute the quantum overlap of this resulting state with a generic intermediate state $|\phi_{n,m}\rangle$. For a non-zero overlap to exist, the intermediate state must contain exactly 1 electron and 1 phonon ($n = 1$, $m = 1$), meaning its bra form is constructed as $\langle\phi_{n,m}| = \langle 0|c_{\mathbf{p}+\mathbf{q}',\sigma'}a_{\mathbf{q}'}$. Taking the inner product systematically isolates the exact phonon mode that was emitted:

$$\begin{aligned} \langle\phi_{n,m}|V_1|\phi_0\rangle &= \left(\sum_{\mathbf{q}',\sigma'}\langle 0|c_{\mathbf{p}+\mathbf{q}',\sigma'}a_{\mathbf{q}'}\right)\left(\sum_{\mathbf{q}} M_{\mathbf{q}}(a^\dagger_{\mathbf{q}} + a_{-\mathbf{q}})c^\dagger_{\mathbf{p}+\mathbf{q},\sigma}|0\rangle\right) \\ &= \sum_{\sigma',\mathbf{q}'} \delta_{\sigma,\sigma'}\delta_{\mathbf{q},\mathbf{q}'}M_{\mathbf{q}} = M_{\mathbf{q}}. \end{aligned}$$

Finally, substituting this squared matrix element back into the formal second-order energy formula, we must evaluate the corresponding energy denominator. The initial unperturbed energy is simply the kinetic energy of the single electron, $E_0^{(\text{unp.})} = \epsilon_{\mathbf{p}}$, and the intermediate unperturbed energy includes both the shifted electron and the newly emitted phonon, $E_{n,m}^{(\text{unp.})} = \epsilon_{\mathbf{p}+\mathbf{q}} + \hbar\omega_{\mathbf{q}}$. Thus, the second-order correction is:

$$E_2 = \frac{M_{\mathbf{q}}^2}{-\hbar\omega_{\mathbf{q}} + (\epsilon_{\mathbf{p}} - \epsilon_{\mathbf{p}+\mathbf{q}})}. \tag{2.1.1}$$

This crucial formula explicitly defines the effective energy shift that the virtual phonon cloud induces on a single isolated fermion.

2.1.3 | Two Body Problem

To rigorously study the effective electron-electron interactions mediated by phonons, we must transition from a single-particle picture to an initial unperturbed state $|\phi_0\rangle$ containing at least two particles. We define this two-fermion ground state as:

$$|\phi_0\rangle = c^\dagger_{\mathbf{p},\alpha} c^\dagger_{\mathbf{p}',\alpha'} |0\rangle.$$

As established previously, the first-order perturbation theory correction trivially vanishes because the interaction Hamiltonian strictly creates or destroys a single phonon, ensuring orthogonality with the initial phonon-free state. We therefore proceed directly to evaluate the second-order energy correction, which requires computing the transition matrix elements connecting our initial two-particle state to the virtual intermediate states:

$$\langle \phi_{n,m} | V_1 | \phi_0 \rangle \quad \text{and} \quad \langle \phi_{2,1} | V_1 | \phi_0 \rangle.$$

Applying the electron-phonon interaction operator V_1 to our initial state allows the fermionic destruction operator $c_{\mathbf{k},\sigma}$ within V_1 to contract with either of the two initial electrons (possessing momenta $\mathbf{p}$ or $\mathbf{p}'$):

$$\begin{aligned} V_1|\phi_0\rangle &= \sum_{\mathbf{q}} M_{\mathbf{q}} (a^\dagger_{-\mathbf{q}} + a_{\mathbf{q}}) \sum_{\mathbf{k},\sigma} c^\dagger_{\mathbf{k}+\mathbf{q},\sigma} c_{\mathbf{k},\sigma} c^\dagger_{\mathbf{p},\alpha} c^\dagger_{\mathbf{p}',\alpha'} |0\rangle \\ &= \sum_{\mathbf{q}} M_{\mathbf{q}} (a^\dagger_{-\mathbf{q}} + a_{\mathbf{q}}) \left(c^\dagger_{\mathbf{p}+\mathbf{q},\alpha} c^\dagger_{\mathbf{p}',\alpha'} + c^\dagger_{\mathbf{p},\alpha} c^\dagger_{\mathbf{p}'+\mathbf{q},\alpha'} \right) |0\rangle. \end{aligned}$$

This branching leads to two distinct physical scattering processes:

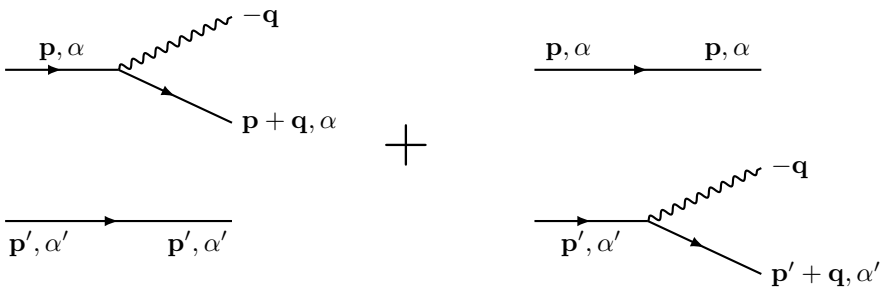


Figure 2.4: Visual representation of the intermediate state created by the interaction operator V_1 acting on the two-electron vacuum, with time flowing horizontally from left to right. The state is a coherent quantum superposition (denoted by the + sign) of two distinct physical processes. Left: The first electron scatters by emitting a phonon $-\mathbf{q}$, while the second electron propagates freely as a spectator. Right: The first electron acts as the free spectator, while the second electron scatters by emitting a phonon $\mathbf{q}$.

(?) Coherent quantum superposition?

These two diagrams correspond precisely to the two possible intermediate states containing exactly one virtual phonon, which can be emitted by either the first electron or the second electron. The bra forms of these specific intermediate states that yield a non-zero overlap are given by:

$$\langle \phi_{2,1}| = \langle 0 | a_{-\mathbf{q}} c_{\mathbf{p}+\mathbf{q},\alpha} c_{\mathbf{p}',\alpha'},$$

$$\langle \tilde{\phi}_{2,1}| = \langle 0 | a_{\mathbf{q}} c_{\mathbf{p},\alpha} c_{\mathbf{p}'-\mathbf{q},\alpha'}.$$

We now substitute these transition matrix elements into the standard second-order perturbation formula. The expectation values of the paired creation and annihilation operators simply trace out to unity, allowing us to factor out the squared electron-phonon matrix element $M_{\mathbf{q}}^2$:

$$E_2' = \left[\frac{\langle \phi_0 | V_1 | \phi_{2,1} \rangle \langle \phi_{2,1} | V_1 | \phi_0 \rangle}{E_0 - E_{2,1}^{(\text{unp.})}} + \frac{\langle \phi_0 | V_1 | \tilde{\phi}_{2,1} \rangle \langle \tilde{\phi}_{2,1} | V_1 | \phi_0 \rangle}{E_0 - \tilde{E}_{2,1}^{(\text{unp.})}} \right]$$
$$= M_{\mathbf{q}}^2 \left(\frac{1}{E_0 - E_{2,1}^{(\text{unp.})}} + \frac{1}{E_0 - \tilde{E}_{2,1}^{(\text{unp.})}} \right).$$

By evaluating this algebraic sum, we can determine whether the resulting energy shift represents a repulsive or an attractive effective interaction between the electrons, a property completely dictated by the sign of the unperturbed energy denominators.

Defining the initial unperturbed energy and the intermediate unperturbed energies for these two distinct exchange processes, we have:

$$E_0 = \epsilon_{\mathbf{p}} + \epsilon_{\mathbf{p}'},$$
$$E_{2,1}^{(\text{unp.})} = \epsilon_{\mathbf{p}+\mathbf{q}} + \epsilon_{\mathbf{p}'} + \hbar\omega_{-\mathbf{q}},$$
$$\tilde{E}_{2,1}^{(\text{unp.})} = \epsilon_{\mathbf{p}'-\mathbf{q}} + \epsilon_{\mathbf{p}} + \hbar\omega_{\mathbf{q}}.$$

Substituting these explicit energies into our second-order expression, the energy shift E_2' becomes:

$$E_2' = M_{\mathbf{q}}^2 \left(\frac{1}{\epsilon_{\mathbf{p}} - \epsilon_{\mathbf{p}+\mathbf{q}} - \hbar\omega_{\mathbf{q}}} + \frac{1}{\epsilon_{\mathbf{p}'} - \epsilon_{\mathbf{p}'-\mathbf{q}} - \hbar\omega_{\mathbf{q}}} \right).$$

To simplify this result, we apply the physical limit of a small momentum exchange $\mathbf{q}$. We assume the relevant scattering processes keep the electrons tightly confined close to the Fermi surface (within the heavily populated Fermi sea), which imposes the approximate kinematic relation:

$$\epsilon_{\mathbf{p}} - \epsilon_{\mathbf{p}+\mathbf{q}} \simeq \epsilon_{\mathbf{p}'-\mathbf{q}} - \epsilon_{\mathbf{p}'}.$$

Applying this small-$\mathbf{q}$ approximation, the effective energy shift neatly reduces to the Fröhlich interaction form:

$$E_2' \simeq M_{\mathbf{q}}^2 \left(\frac{2\hbar\omega_{\mathbf{q}}}{(\epsilon_{\mathbf{p}+\mathbf{q}} - \epsilon_{\mathbf{p}})^2 - (\hbar\omega_{\mathbf{q}})^2} \right). \tag{2.1.2}$$

This pivotal result reveals that the effective phonon-mediated interaction can dynamically manifest as either attractive or repulsive, depending entirely on the sign of the energy denominator.

Attractive Interaction and Isotope Effect

We now restrict our focus to the low-temperature regime, specifically when $T \ll T_D$, where $T_D \sim \omega_D$ is the characteristic Debye temperature of the crystal lattice. In this regime, the electrons lack the thermal energy necessary to scatter far from the Fermi surface. Consequently, the kinetic energy difference between the initial and scattered electron states is constrained to be much smaller than the typical phonon energy (the Debye energy):

$$|\epsilon_{\mathbf{k}+\mathbf{q}} - \epsilon_{\mathbf{k}}| \ll \hbar\omega_D.$$

Under this stringent condition, the denominator in Equation 2.1.2 is overwhelmingly dominated by the phonon energy squared term, $-(\hbar\omega_{\mathbf{q}})^2$, ensuring it naturally becomes negative. The effective interaction therefore simplifies to a constant, weakly attractive potential:

$$E_2' \simeq \frac{-|M_{\mathbf{q}}|^2}{\hbar\omega_D}. \tag{2.1.3}$$

Recall from our fundamental coupling definitions that the squared electron-phonon matrix element, $|M_{\mathbf{q}}|^2$, is inversely proportional to the mass of the lattice ions (M_{ion}). Therefore, the overall strength of the attractive interaction scales directly as:

$$E_2' \propto \frac{1}{M_{\text{ion}}}.$$

This simple inverse proportionality provides a profound microscopic explanation for the experimentally observed **isotope effect** in conventional superconductors, where the critical transition temperature is found to vary predictably with the specific mass of the isotopes that comprise the crystal lattice.

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Remark. *While our derivation treats V_1 as a small perturbation within a simplified cubic lattice, this is not always universally applicable. Because the matrix element $M_{\mathbf{q}}$ scales inversely with the ion mass, the effective electron-phonon interaction can become exceedingly strong in materials composed of very light ions. In such strong-coupling regimes, the perturbative approach breaks down, necessitating more advanced non-perturbative methods (such as Eliashberg theory). Furthermore, real materials often possess complex crystal geometries far beyond a simple cubic structure; these specific geometric details fundamentally alter the phonon spectrum and, consequently, the precise structure of the effective electron-electron interaction.*

The BCS Effective Potential

For the specific purposes of successfully developing the macroscopic BCS theory (Step 1 of the paradigm), we treat this phonon-mediated interaction as a separable, strictly attractive potential. We completely drop its explicit momentum dependence in favor of a constant effective coupling strength, phenomenologically denoted as $-|g_{\text{eff}}|^2$.

Furthermore, while the complete interaction Hamiltonian realistically includes the residual screened Coulomb repulsion (the Thomas-Fermi term), the fundamental assumption of BCS theory is that the attractive phonon-mediated part robustly dominates for electrons near the Fermi surface. Thus, we neglect the residual electrostatic repulsion and approximate the total effective electron-electron potential as purely attractive:

$$V_{e-e,\mathbf{k},\mathbf{k}'} \simeq -|g_{\text{eff}}|^2 + \cancel{V_{\text{Thomas-Fermi}}} \simeq -|g_{\text{eff}}|^2.$$

It is crucial to emphasize that while this attractive energy shift $\Delta E = -|g_{\text{eff}}|^2$ (derived by discarding scattering processes with energy exchanges far beyond the Debye cutoff ω_D) is exceedingly small, it is fundamentally sufficient to destabilize the unperturbed Fermi sea and induce the formation of a bound state between two electrons. Having established the scalar amplitude of this

attractive potential, the immediate next step is to rigorously construct the full operatorial part of the interaction Hamiltonian to be utilized in solving the macroscopic many-body problem.

We are assuming weak interactions, but that is not always the case: the coefficient $M(\mathbf{q})$ is inversely proportional to the mass of the ions in the lattice, and thus the interaction can be very strong in the case of light ions. In this case, we need to use a non-perturbative approach.

One has to pay attention also to the geometry of the lattice, since our treatment was based on a cubic lattice, but in general the lattice can be more complicated. The geometry of the lattice can affect the phonon spectrum and thus the effective interaction between electrons.

We got $\Delta E = -|g_{eff}|^2$ for phonons with $\omega < \omega_D$ (discarding processes going to far away from the Fermi surface). This is a very small energy, but it is enough to create a bound state of two electrons. Now we have to find the operatorial part of the Hamiltonian, which is the part that will be used in the many-body problem.

2.1.4 | Effective Hamiltonian and Two-Body Ground State

In the second quantization formalism, the interacting part of the many-body Hamiltonian, H_{int}, capturing this fundamental scattering process between pairs of fermions, can be modeled as follows:

$$H_{\text{int}} = \frac{1}{2V} \sum_{\mathbf{k},\mathbf{k}',\mathbf{q}} \sum_{\sigma,\sigma'} V(\mathbf{k}',\mathbf{k}) c^\dagger_{\mathbf{k}'+\frac{\mathbf{q}}{2},\sigma} c^\dagger_{-\mathbf{k}'+\frac{\mathbf{q}}{2},\sigma'} c_{-\mathbf{k}+\frac{\mathbf{q}}{2},\sigma'} c_{\mathbf{k}+\frac{\mathbf{q}}{2},\sigma}, \tag{2.1.4}$$

where V is the macroscopic volume of the system, and $c_{\mathbf{k},\sigma}$ and $c^\dagger_{\mathbf{k},\sigma}$ are respectively the fermionic annihilation and creation operators for an electron possessing momentum $\mathbf{k}$ and spin projection σ. The scattering amplitude is governed by the matrix element $V(\mathbf{k}',\mathbf{k})$. We are summing over all possible incoming and outgoing momenta (and also over all feasible values of $\mathbf{q}$), as well as over the spin indices σ and σ'. The factor of $1/2$ is included to avoid double counting of identical scattering processes.

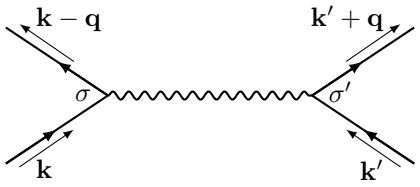


Figure 2.5: Feynman diagram representing the effective electron-electron interaction mediated by the virtual exchange of a phonon. Two incoming fermions with momenta $\mathbf{k}$ and $\mathbf{k}'$ scatter into two outgoing fermions with momenta $\mathbf{k}-\mathbf{q}$ and $\mathbf{k}'+\mathbf{q}$, exchanging a momentum $\mathbf{q}$ and preserving their respective spins σ and σ'.

Notice how phonons have disappeared from the final expression for the interaction Hamiltonian (2.1.4). This is because we have integrated out the phonon degrees of freedom, leaving only an effective interaction between the electrons. The phonons are still present in the system, but they are now treated as virtual excitations that mediate the interaction between the electrons. Our goal is now to "solve" the many-body problem of interacting fermions with this effective interaction, which is a highly non-trivial task. Thus we will limit our analysis to the simplest possible model that still captures the essential physics of the problem: the ground state of a system with 2 electrons interacting via the effective potential $V(\mathbf{k}',\mathbf{k})$ in the presence of a filled Fermi sea of non-interacting

electrons. This is the original problem solved by Cooper, and it is the starting point for the BCS theory.

We are then trying to compute the following wavefunction in the center of mass frame:

$$\Psi(\mathbf{r}_1, \sigma_1; \mathbf{r}_2, \sigma_2) = e^{i\mathbf{k}_{cm}\cdot\mathbf{R}_{cm}}\varphi(\mathbf{r}_1 - \mathbf{r}_2)\phi(\sigma_1, \sigma_2),$$

where we assumed translational invariance and separated the center of mass motion from the relative motion. We were able to factorize the wavefunction into a plane wave for the center of mass motion, a spatial wavefunction for the relative motion, and a spin wavefunction, since in the hamiltonian the spin and spatial degrees of freedom are decoupled (but there may be cases in which we have some sort of spin-orbit coupling, which would couple the spin and spatial degrees of freedom).

Considering the spin wavefunction, we have two possibilities: the singlet state (antisymmetric) and the triplet state (symmetric). The Pauli exclusion principle requires that if we are requiring local pairing in the same state, then we can only accept states like $|\uparrow\downarrow\rangle$ and $|\downarrow\uparrow\rangle$ (plus linear combinations). Thus we have

$$\text{- singlet:} \quad \phi(\sigma_1, \sigma_2) = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle),$$

$$\text{- triplet:} \quad \phi(\sigma_1, \sigma_2) = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle),$$

but then the triplet is unfavorable by energy considerations...

We now what the function to be antisymmetric for exchange of the two electrons, and the spin part is antisymmetric, so the spatial part must be symmetric. This means that the spatial wavefunction must depend only upon the distance between the two electrons, $\varphi(\mathbf{r}_1 - \mathbf{r}_2) = \varphi(r)$. Thus the problem now depends only on the relative distance between the two electrons, which is a simpler treatment. Then we can expand in Fourier modes the spatial wavefunction:

$$\varphi(r) = \sum_{\mathbf{k}} \varphi_{\mathbf{k}} e^{i\mathbf{k}\cdot(\mathbf{r}_1 - \mathbf{r}_2)}.$$

and we also need to require that $\varphi_{\mathbf{k}} = \varphi_{-\mathbf{k}}$ because the spatial wavefunction is symmetric. We can then write the Schrödinger equation for the relative motion of the two electrons in momentum space:

$$E\Psi = (H_0 + H_{int})\Psi, \quad \Psi = \sum_{\mathbf{k}} \varphi_{\mathbf{k}} e^{i\mathbf{k}\cdot(\mathbf{r}_1 - \mathbf{r}_2)}\phi(\sigma_1, \sigma_2),$$

In order to find the functional form of the wavefunction φ, we need to compute the following matrix element:

$$\langle \Psi_{\mathbf{k}} | H | \Psi \rangle ,$$

where $|\Psi\rangle = \sum_{\mathbf{k}} \varphi_{\mathbf{k}} |\Psi_{\mathbf{k}}\rangle$ and

$$|\Psi_{\mathbf{k}}\rangle = \frac{1}{\sqrt{2}} \left(c^\dagger_{\mathbf{k}\uparrow} c^\dagger_{-\mathbf{k}\downarrow} - c^\dagger_{-\mathbf{k}\uparrow} c^\dagger_{\mathbf{k}\downarrow} \right) |0\rangle .$$

Now we split the computation into two parts: the free non interacting part and the interaction part. The kinetic part is straightforward:

$$H_0 |\Psi\rangle = \left(\frac{\hat{p}_1^2}{2m} + \frac{\hat{p}_2^2}{2m} \right) \left(\sum_{\mathbf{k}}{}' \varphi_{\mathbf{k}} e^{i\mathbf{k}\cdot(\mathbf{r}_1 - \mathbf{r}_2)}\phi(\sigma_1, \sigma_2) \right) = 2 \sum_{\mathbf{k}}{}' \epsilon_{\mathbf{k}} \varphi_{\mathbf{k}} e^{i\mathbf{k}\cdot(\mathbf{r}_1 - \mathbf{r}_2)}\phi(\sigma_1, \sigma_2).$$

For the interaction part, we have to compute the following:

$$H_{int}\left|\Psi\right\rangle = -|g_{eff}|^2 \sum_{\sigma_1,\sigma_2} \sum_{\mathbf{k}_1,\mathbf{k}_2} {\sum_{\mathbf{q}}}' c^\dagger_{\mathbf{k}_1+\mathbf{q},\sigma_1} c^\dagger_{\mathbf{k}_2-\mathbf{q},\sigma_2} c_{\mathbf{k}_1,\sigma_1} c_{\mathbf{k}_2,\sigma_2} \left({\sum_{\mathbf{k}}}' \varphi_{\mathbf{k}} e^{i\mathbf{k}\cdot(\mathbf{r}_1-\mathbf{r}_2)} \right) \left(c^\dagger_{\mathbf{k}\uparrow} c^\dagger_{-\mathbf{k}\downarrow} - c^\dagger_{-\mathbf{k}\uparrow} c^\dagger_{\mathbf{k}\downarrow} \right) \left|0\right\rangle ,$$

which after some algebra and the summations resolved over the deltas, we get:

$$(\sigma_{1,\uparrow}, \sigma_{2,\downarrow}) = -|g_{eff}|^2 {\sum_{\mathbf{q}}}' {\sum_{\mathbf{k}}}' \varphi_{\mathbf{k}} \left(c^\dagger_{\mathbf{k}+\mathbf{q},\uparrow} c^\dagger_{\mathbf{k}-\mathbf{q},\downarrow} - c^\dagger_{-\mathbf{k}+\mathbf{q},\uparrow} c^\dagger_{\mathbf{k}+\mathbf{q},\downarrow} \right) \left|0\right\rangle .$$

Now if we compute the complete matrix element, we get the following equation for the coefficients $\varphi_{\mathbf{k}}$:

$$\left\langle \Psi_{\mathbf{k}} \right| E \left|\Psi\right\rangle = E\varphi_{\mathbf{k}},$$

such that:

$$\left\langle \Psi_{\mathbf{k}} \right| H_0 + H_{int} \left|\Psi\right\rangle = 2\epsilon_{\mathbf{k}}\varphi_{\mathbf{k}} + \cdots = \langle 0| \left(c_{-\mathbf{k},\uparrow} c_{\mathbf{k},\downarrow} - c_{\mathbf{k},\downarrow} c_{-\mathbf{k},\uparrow} \right) \times {\sum_{\mathbf{k}'}}' \varphi_{\mathbf{k}'} e^{-i\mathbf{k}'\cdot(\mathbf{r}_1-\mathbf{r}_2)}$$
$$\times {\sum_{\mathbf{q}}}' (-|g_{eff}|^2) \left(c^\dagger_{\mathbf{k}'+\mathbf{q},\uparrow} c^\dagger_{-\mathbf{k}'-\mathbf{q},\downarrow} - c^\dagger_{-\mathbf{k}'+\mathbf{q},\uparrow} c^\dagger_{\mathbf{k}'-\mathbf{q},\downarrow} \right) \left|0\right\rangle .$$

[...]

$$\implies (E - 2\epsilon_{\mathbf{k}})\varphi_{\mathbf{k}} = -|g_{eff}|^2 {\sum_{\mathbf{k}'}}' \varphi_{\mathbf{k}'},$$

such that now we have a closed equation for the coefficients $\varphi_{\mathbf{k}}$. We can now sum over $\mathbf{k}$ to get:

$$C = {\sum_{\mathbf{k}'}}' \varphi_{\mathbf{k}'} = -|g_{eff}|^2 {\sum_{\mathbf{k}}}' \frac{C}{E - 2\epsilon_{\mathbf{k}}} = -|g_{eff}|^2 C {\sum_{\mathbf{k}}}' \frac{1}{E - 2\epsilon_{\mathbf{k}}}.$$

If we now go to the continuum limit, we can replace the sum over $\mathbf{k}$ with an integral over the density of states $g(\epsilon)$:

$$\sum_{\mathbf{k}} \xrightarrow{cont.lim.} \int_0^{\hbar\omega_D} d\epsilon\, g(\epsilon),$$

then

$$1 = -|g_{eff}|^2 \int_0^{\hbar\omega_D} d\epsilon \, \frac{g(\epsilon)}{E - 2\epsilon} \sim -|g_{eff}|^2 g(E_F) \int_0^{\hbar\omega_D} d\epsilon \, \frac{1}{E - 2\epsilon} = -|g_{eff}|^2 g(E_F) \frac{1}{2} \ln\left(\frac{E - 2\hbar\omega_D}{E} \right),$$

$$\frac{1}{\lambda} = -\frac{1}{2} \frac{1}{|g_{eff}|^2 g(E_F)} = \ln\left(\frac{E - 2\hbar\omega_D}{E} \right) \implies E = -2\hbar\omega_D e^{-2/\lambda},$$

$$1 - e^{\frac{1}{\lambda}} = \frac{2\hbar\omega_D}{E} \implies E = \frac{2\hbar\omega_D}{1 - e^{1/\lambda}},$$

which means that $E < 0$: the two electrons form a bound state, which is the Cooper pair. The binding energy is exponentially small in the coupling constant λ, which is a measure of the strength of the effective interaction. This result shows that even an arbitrarily weak attractive interaction can lead to the formation of bound pairs of electrons, which is the key insight of the BCS theory of superconductivity.

We can also compute a limit for $\hbar\omega_D \gg E$ then

$$\frac{1}{\lambda} \sim \log\left(\frac{2\hbar\omega_D}{|E|} \right) \implies |E| \sim 2\hbar\omega_D e^{-|g_{eff}|^2 g(E_F)}.$$

This is the famous result of Cooper, which shows that the binding energy of the Cooper pair is exponentially small in the coupling constant λ. This result is crucial for understanding the formation of Cooper pairs and the onset of superconductivity in metals.

Let us reason a bit on the temperature scale: the fermi temperature for a typical metal is on the order of $T_F \sim 10^4\,\mathrm{K}$, and $g_{eff} \sim \frac{10^4}{10^2} \sim 10^2$ thus the energy scale $E \sim \frac{10^2}{10^2} = 1K$, since we are at second order in perturbation theory, and this is the typical scale of the critical temperature for superconductivity. This shows that the BCS theory provides a natural explanation for the low critical temperatures observed in conventional superconductors. There may be other bound states with higher energy, but our perturbative analysis does not allow us to access them.

[...]

[...]

2.2 | From Microscopic to Mesoscopic Descriptions

[...]

TODO: erio notes...

2.2.1 | The Effective Attractive Potential

To capture the essential physics of the Fröhlich mechanism, specifically that the phonon-mediated attraction only overcomes Coulomb repulsion for electrons energetically close to the Fermi surface, the effective attractive potential is explicitly modeled by a piecewise function:

$$V(\mathbf{k}', \mathbf{k}) = \begin{cases} -g_{\text{e,ph}}^2 & \text{if } |\epsilon_k - \mu| < \hbar\omega_D \text{ and } |\epsilon_{k'} - \mu| < \hbar\omega_D, \\ 0 & \text{elsewhere,} \end{cases} \tag{2.2.1}$$

where $g_{\text{e,ph}}$ represents the effective coupling constant between the electrons and phonons, μ is the chemical potential of the electron gas (recalling that in the low-temperature limit $T \to 0$, the chemical potential intimately approaches the Fermi energy, $\mu(T) \xrightarrow{T\to 0} E_F$), ϵ_k is the non-interacting kinetic energy of the fermion, and ω_D is the characteristic Debye frequency of the ionic lattice.

By analyzing the explicit energy bounds in the expression for the potential (2.2.1), we clearly see a profound physical restriction: only those electrons located within a remarkably narrow energy shell surrounding the Fermi sphere are permitted to interact attractively with each other.

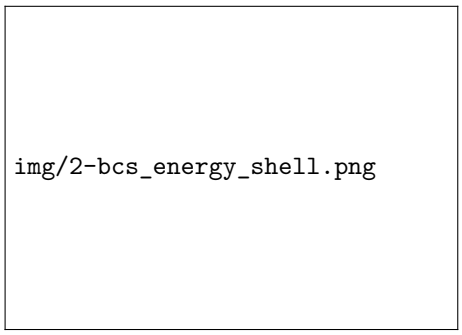


Figure 2.6: The model of the potential $V(\mathbf{k}', \mathbf{k})$ for superconductors is given by (2.2.1) and it involves just the fermions with energies ϵ_k and $\epsilon_{k'}$ residing strictly within a tiny energy shell of width $2\hbar\omega_D$ centered around the Fermi energy.

To express this restricted interaction more compactly, it is highly advantageous to introduce a dimensionless cut-off function $w_k = \Theta(\hbar\omega_D - |\epsilon_k - \mu|)$, where Θ denotes the standard Heaviside step function. Utilizing this notation, the piecewise potential (2.2.1) can be elegantly rewritten in a strictly factorized form:

$$V(\mathbf{k}', \mathbf{k}) = -g_{\text{e,ph}}^2 w_k w_{k'}. \tag{2.2.2}$$

This mathematically represents the localized attractive potential acting exclusively between fermions scattered around the Fermi energy in a superconductor. Note that this specific theoretical form of the potential is entirely *separable*; the dependence on the incoming momentum $\mathbf{k}$ and the outgoing momentum $\mathbf{k}'$ factors into independent terms, a property that will dramatically simplify the subsequent many-body calculations.

2.2.2 | Ultracold Fermi Gases

As previously mentioned, the BCS theoretical framework can also be employed to study systems of ultracold Fermi atoms. In these highly controlled systems, the effective interparticle interaction

possesses a very short range compared to the average interparticle distance. Because of this separation of scales, it is highly convenient to model the interaction using a separable potential in momentum space:

$$V(\mathbf{k}', \mathbf{k}) = v_0 w_k w_{k'},$$

where $w_k = \Theta(k_0 - k)$ utilizes the Heaviside step function to introduce a strict *ultraviolet momentum cutoff* k_0, and v_0 represents the bare interaction strength. Note that formally, this separable potential is structurally very similar to the one utilized for modeling electron-phonon interactions in superconductors.

Scattering Theory

For a genuinely short-range potential in position space, the corresponding cutoff in momentum space must be taken to be very large, meaning $k_0 \to \infty$. We thus need to understand how to appropriately treat this divergent limit mathematically. The core physical idea is that as the cutoff k_0 goes to infinity, the bare interaction parameter v_0 must simultaneously change with k_0 in a precise manner in order to keep the physically observable **scattering length** a strictly constant.

To rigorously establish this relationship, we invoke the quantum theory of scattering.[1] We know that the elastic scattering amplitude $f(\mathbf{k}', \mathbf{k}) \equiv f(\mathbf{k}' \leftarrow \mathbf{k})$ for scattering from an initial momentum $\mathbf{k}$ to a final momentum $\mathbf{k}'$ can be obtained directly from the T-matrix, $T(\mathbf{k}', \mathbf{k}, z)$. This T-matrix satisfies the fundamental *Lippmann-Schwinger integral equation*:

$$T(\mathbf{k}', \mathbf{k}, z) = V(\mathbf{k}', \mathbf{k}) + \int \frac{\mathrm{d}^3\mathbf{k}''}{(2\pi)^3} \frac{V(\mathbf{k}', \mathbf{k}'')}{z - \frac{(k'')^2}{2m_r}} T(\mathbf{k}'', \mathbf{k}, z),$$

where m_r is the reduced mass of the colliding two-body system, while z is the complex energy variable. For a gas of identical fermions of mass m, the reduced mass is simply $m_r = m/2$. The formal relation linking the T-matrix to the physical scattering amplitude is evaluated on-shell:

$$f(\mathbf{k}', \mathbf{k}) = -\frac{m_r}{2\pi\hbar^2} T\left(\mathbf{k}', \mathbf{k}, z = \frac{\hbar^2 k^2}{2m_r} + i0^+\right),$$

where $i0^+$ is an infinitesimal positive imaginary term. In the extreme low-energy limit, the scattering amplitude limits to the negative of the scattering length, implying that:

$$T(0, 0, 0) = \frac{2\pi\hbar^2}{m_r} a. \tag{2.2.3}$$

We can safely ignore the infinitesimal term $i0^+$ because it becomes entirely irrelevant for the principal value of the integral when evaluating at exactly $z = 0$. Here we are translating the physical requirement of a fixed scattering length a into a precise mathematical condition on the bare interaction strength v_0 as a function of the ultraviolet cutoff k_0: it is a microscopic requirement that we have derived directly from mesoscopic scattering theory.

Renormalization of the Bare Coupling Strength

Because our proposed potential is purely separable and takes the form $V(\mathbf{k}', \mathbf{k}) = v_0 w_k w_{k'}$, the integral equation can be easily solved analytically. Substituting the potential yields:

$$T(\mathbf{k}', \mathbf{k}, z) = v_0 w_k w_{k'} + v_0 w_{k'} \int \frac{\mathrm{d}^3\mathbf{k}''}{(2\pi)^3} \frac{w_{k''}}{z - \frac{(k'')^2}{2m_r}} T(\mathbf{k}'', \mathbf{k}, z).$$

[1] For a comprehensive treatment, see, e.g., John R. Taylor, *Scattering Theory: The Quantum Theory of Nonrelativistic Collisions* (Dover Books).

We can now employ the separable ansatz $T(\mathbf{k}', \mathbf{k}, z) = T(z) w_k w_{k'}$: we can do this since if we take $\mathbf{k}$ and $\mathbf{k}'$ to be out of the annulus defined by the cutoff, then $T(\mathbf{k}', \mathbf{k}, z) = 0$ and the equation is trivially satisfied. Otherwise, if only one of $\mathbf{k}$ or $\mathbf{k}'$ is outside the annulus, then $T(\mathbf{k}', \mathbf{k}, z) = 0$ because it has to be invariant under the exchange of $\mathbf{k}$ and $\mathbf{k}'$. Thus, we factor out the momentum dependence and obtain a simple algebraic equation for the scalar function $T(z)$:

TODO: explain better the ansatz

$$T(z) = v_0 + v_0 \int \frac{\mathrm{d}^3\mathbf{k}''}{(2\pi)^3} \frac{w_{k''}^2}{z - \frac{(k'')^2}{2m_r}} T(z).$$

This equation is readily solved for $T(z)$, yielding:

$$T(z) = \frac{v_0}{1 - v_0 \int \frac{\mathrm{d}^3\mathbf{k}''}{(2\pi)^3} \frac{w_{k''}^2}{z - \frac{(k'')^2}{2m_r}}}.$$

Setting $z = 0$ to connect with our low-energy scattering length boundary condition provides:

$$T(0) = \frac{1}{\frac{1}{v_0} + \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} \frac{w_k^2}{\frac{k^2}{2m_r}}}.$$

Equating this result to the previously derived kinematic definition for the scattering length, $T(0,0,0) = \frac{4\pi\hbar^2}{m} a$ (where we have substituted $m_r = m/2$ w.r.t. (2.2.3)), we obtain:

$$\frac{4\pi\hbar^2}{m} a = \frac{1}{\frac{1}{v_0} + \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} \frac{w_k^2}{\frac{k^2}{2m_r}}}.$$

Remark. *Since we used (2.2.3) to define the scattering length, this relation is valid in the low-energy limit, where the scattering amplitude is independent of the incoming and outgoing momenta. In this limit, the scattering length is a universal quantity that characterizes the low-energy scattering properties of the system, independent of the specific details of the interaction potential.*

By isolating the inverse bare coupling strength, we can rewrite this relation as:

$$\frac{1}{v_0} = \frac{m}{4\pi\hbar^2 a} - \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} \frac{w_k^2}{2\epsilon_k}, \tag{2.2.4}$$

where we have explicitly used the free-particle kinetic energy dispersion $\epsilon_k = \hbar^2 k^2 / 2m$. Equation (2.2.4) reveals how the bare interaction strength v_0 must scale intricately with the ultraviolet momentum cutoff k_0 (which is embedded within the definition of w_k) when the macroscopic scattering length a is held strictly fixed. This mathematical behavior is deeply analogous to a *running coupling constant* in quantum field theory. It is critical to note that (2.2.4) is universally valid for any generic separable potential, independent of the specific functional form chosen for the cutoff w_k.

For our specific hard-sphere cutoff model, $w_k = \Theta(k_0 - k)$, the integral can be evaluated analytically in spherical coordinates, yielding:

$$v_0 = \frac{\hbar^2}{m} \frac{1}{\frac{1}{4\pi a} - \frac{k_0}{2\pi^2}}.$$

To analyze the physics of a true zero-range contact interaction, we take the limit as the cutoff $k_0 \to \infty$ while holding a constant (such that the dimensionless parameter $k_0 a \gg 1$). Expanding the expression in this limit gives:

$$v_0 = -\frac{2\pi^2\hbar^2}{m k_0} \left[1 + \frac{\pi}{2 k_0 a}\right].$$

This asymptotic expansion clearly demonstrates that in the limiting case of a perfectly localized contact potential, the bare coupling strength must approach zero from below, $v_0 \to 0^-$, as $k_0 \to \infty$. Notice also that the physical scattering length a appears exclusively in the subleading term on the right-hand side of this limiting expression. Finally, we are not really concerned with the fact of having a momentum cutoff in the interaction, since the physical scattering length is a well-defined quantity that can be measured experimentally and we are not really interested in the microscopic details of the interaction potential, expecially at such high momenta.

2.3 | Grand Canonical Hamiltonian and Ground State

Let us now proceed to study the many-body problem. We will work within the grand canonical ensemble, where the thermodynamic properties of the system are governed by the partition function $\mathcal{Z}$, defined as:

$$\mathcal{Z} = \mathrm{Tr}\left\{e^{-\beta \hat{K}}\right\},$$

where $\hat{K} = \hat{H} - \mu \hat{N}$ is the grand canonical "Hamiltonian", μ is the chemical potential, and $\hat{N}$ is the fermionic number operator.

Notice that the partition function $\mathcal{Z}$ is not an observable: it is a purely mathematical object that encodes the statistical properties of the system, which is related to different microscopic and thermodinamic observables; but it is also a matrix with a number of eigenvalues that grows exponentially with the number of particles in the system: this makes it impossible to compute exactly for a many-body system.

Using the standard second-quantized expression for the interaction, $\hat{K}$ takes the form:

$$\hat{K} = \sum_{\mathbf{k},\sigma} \xi_k c^\dagger_{\mathbf{k},\sigma} c_{\mathbf{k},\sigma} + \frac{1}{2V} \sum_{\mathbf{k},\mathbf{k}',\mathbf{q}} \sum_{\sigma,\sigma'} V(\mathbf{k}',\mathbf{k}) c^\dagger_{\mathbf{k}'+\frac{\mathbf{q}}{2},\sigma} c^\dagger_{-\mathbf{k}'+\frac{\mathbf{q}}{2},\sigma'} c_{-\mathbf{k}+\frac{\mathbf{q}}{2},\sigma'} c_{\mathbf{k}+\frac{\mathbf{q}}{2},\sigma},$$

where we have defined the single-particle kinetic energy relative to the chemical potential as $\xi_k = \epsilon_k - \mu$.[2] For notational convenience, we will write the non-interacting kinetic part of $\hat{K}$ as $\hat{T}$.

We are fundamentally interested in the pairing mechanism between fermions of opposite spin (spin up $\uparrow$ and spin down $\downarrow$), as this is the dominant process in most conventional superconductors. The interaction between identical fermions (same spin) is irrelevant for the pairing mechanism and can thus be safely ignored, since it exerts only a negligible effect on the physical properties of a superconductor. Furthermore, in the specific case of ultracold Fermi gases, the effective interaction is strictly short-ranged, meaning only interactions between up and down fermions need to be considered; the presence of two identical fermions in the same spatial region is strongly suppressed by the Pauli exclusion principle. By restricting the interaction term to couple only opposite spins, and performing the summation over the spin indices σ and σ', we obtain:

$$\hat{K} = \hat{T} + \frac{1}{V} \sum_{\mathbf{k},\mathbf{k}',\mathbf{q}} V(\mathbf{k}',\mathbf{k}) c^\dagger_{\mathbf{k}'+\frac{\mathbf{q}}{2},\uparrow} c^\dagger_{-\mathbf{k}'+\frac{\mathbf{q}}{2},\downarrow} c_{-\mathbf{k}+\frac{\mathbf{q}}{2},\downarrow} c_{\mathbf{k}+\frac{\mathbf{q}}{2},\uparrow}.$$

Keep in mind that all these summations are considering only momenta within a narrow shell around the Fermi surface, for which the effective interaction is non-zero.

Furthermore this hamiltonian has an $U(1)$ symmetry, which is the symmetry associated with the conservation of the number of particles: it commutes with the number operator $\hat{N}$ and there are no terms that create or destroy particles without compensating for it. Furthermore we have a spin-symmetry (called also asymmetric charge) goven by the difference between the number of spin-up and spin-down particles, which is also conserved. This symmetry is associated with the conservation of the total spin of the system, since for any spin-up particle that is created, a spin-down particle is also created, and vice versa (or a spin-up particle isdestroyed).

[2] This follows straightforwardly from the definition of the grand canonical Hamiltonian.

2.3.1 | Mean-Field Approximation

We now proceed in a manner conceptually similar to the treatment of the weakly-interacting Bose gas, with the primary difference being that in the present fermionic case, we anticipate the macroscopic condensation of *pairs* of fermions rather than individual particles.[3] Consider the pair destruction operator factor $c_{-\mathbf{k}+\frac{\mathbf{q}}{2},\downarrow}c_{\mathbf{k}+\frac{\mathbf{q}}{2},\uparrow}$ embedded in the interaction term, which will give rise to an interaction term between **Cooper pairs**. We rewrite it (keeping all momentum and spin indices implicit for brevity) by expanding around its thermodynamic mean:

$$cc = cc - \langle cc\rangle + \langle cc\rangle = \delta(cc) + \langle cc\rangle,$$

where $\delta(cc) = cc - \langle cc\rangle$ represents the deviation (or fluctuation) of the operator cc from its expectation value $\langle cc\rangle$. We will assume this deviation to be a small quantity and apply the exact same decomposition to the pair creation factor $c^\dagger c^\dagger$. Substituting these back into the interaction and systematically neglecting the second-order fluctuation term of order $\delta(c^\dagger c^\dagger)\delta(cc)$, the four-operator interaction is linearized into a bilinear form:

$$\begin{aligned} c^\dagger c^\dagger cc &= \delta(c^\dagger c^\dagger)\langle cc\rangle + \langle c^\dagger c^\dagger\rangle\delta(cc) + \langle c^\dagger c^\dagger\rangle\langle cc\rangle \\ &= c^\dagger c^\dagger\langle cc\rangle + \langle c^\dagger c^\dagger\rangle cc - \langle c^\dagger c^\dagger\rangle\langle cc\rangle. \end{aligned}$$

We have already made an argument to exploit the anti-alignement of the spins of the two fermions in a Cooper pair, which is a fundamental property of the BCS theory. This allows us to restrict our attention to the spin-up and spin-down components of the fermionic operators, as we have done in the above expressions. But we have to still take into account the momentum of the pairs, which is represented by the center-of-mass momentum $\mathbf{q}$. Because the macroscopic condensation of pairs is physically expected to occur exactly at zero center-of-mass momentum, the expectation values are non-zero only for $\mathbf{q} = 0$. Thus, we can rigorously set:

TODO: add figure of opposing momentum fermions in the annulus as only allowed to interact to form a Cooper pair with zero center-of-mass momentum

$$\begin{aligned} \langle c_{-\mathbf{k}+\frac{\mathbf{q}}{2},\downarrow}c_{\mathbf{k}+\frac{\mathbf{q}}{2},\uparrow}\rangle &= \delta_{\mathbf{q},0}A_{\mathbf{k}}\,, \\ \langle c^\dagger_{\mathbf{k}'+\frac{\mathbf{q}}{2},\uparrow}c^\dagger_{-\mathbf{k}'+\frac{\mathbf{q}}{2},\downarrow}\rangle &= \delta_{\mathbf{q},0}A^*_{\mathbf{k}'}\,, \end{aligned}$$

where $A_{\mathbf{k}}$ is a function of the momentum $\mathbf{k}$ that remains to be determined. Substituting this mean-field ansatz, the grand canonical Hamiltonian simplifies to:

$$\hat{K} = \hat{T} + \frac{1}{V}\sum_{\mathbf{k},\mathbf{k}'} V(\mathbf{k}',\mathbf{k})\left[c^\dagger_{\mathbf{k}',\uparrow}c^\dagger_{-\mathbf{k}',\downarrow}A_{\mathbf{k}} + c_{-\mathbf{k},\downarrow}c_{\mathbf{k},\uparrow}A^*_{\mathbf{k}'}\right] - \frac{1}{V}\sum_{\mathbf{k},\mathbf{k}'} V(\mathbf{k}',\mathbf{k})A^*_{\mathbf{k}'}A_{\mathbf{k}}.$$

To render this expression more physically transparent, we define the *momentum-dependent superconducting* **gap parameter** $\Delta_{\mathbf{k}}$ and its complex conjugate:

$$\Delta_{\mathbf{k}'} = -\frac{1}{V}\sum_{\mathbf{k}} V(\mathbf{k}',\mathbf{k})A_{\mathbf{k}}, \quad \Delta^*_{\mathbf{k}} = -\frac{1}{V}\sum_{\mathbf{k}'} V(\mathbf{k}',\mathbf{k})A^*_{\mathbf{k}'}. \tag{2.3.1}$$

Note that these definitions are perfectly consistent because any time-reversal invariant potential satisfies the symmetry $V(\mathbf{k},\mathbf{k}') = V(\mathbf{k}',\mathbf{k})^*$. With these definitions, one obtains a bilinear Hamiltonian:

$$\hat{K} = \hat{T} - \sum_{\mathbf{k}}\left[\Delta_{\mathbf{k}}c^\dagger_{\mathbf{k},\uparrow}c^\dagger_{-\mathbf{k},\downarrow} + \Delta^*_{\mathbf{k}}c_{-\mathbf{k},\downarrow}c_{\mathbf{k},\uparrow}\right] + \sum_{\mathbf{k}} A^*_{\mathbf{k}}\Delta_{\mathbf{k}}.$$

[3] Since fermionic wavefunctions are antisymmetric, when we compute expectation values of fermionic operators we are taking a tensor product of something foundamentally antisymmetric ($c\,|\psi_{GS}\rangle$) and something foundamentally symmetric ($\langle\psi_{GS}|$): this makes the computation of expectation values more complex. We cannot take the expectation value of fermionic operators directly.

Notice that the last term is a constant energy offset that does not depend on the fermionic operators. Usually one discards such terms, but in this case it is important to keep it because it is a function of the gap parameter $\Delta_{\mathbf{k}}$ and thus it will contribute to the self-consistent equation for the gap. We will assume anyway that the Ground State solution will be independent of such constant term, and we will see that this is indeed the case.

The Gaussian Problem we have obtained is a quadratic Hamiltonian in the fermionic operators, which can be solved exactly. The only unknowns are the gap parameters $\Delta_{\mathbf{k}}$ and $\Delta^*_{\mathbf{k}}$, which will be determined self-consistently by solving the gap equation (2.3.1).

2.3.2 | Diagonalization of the Hamiltonian

To elegantly diagonalize this bilinear Hamiltonian, it is highly convenient to express it in a matrix representation using **Nambu spinors**. By leveraging the fundamental fermionic anti-commutation relations, we can rewrite the kinetic energy term for the spin-down particles as

$$\xi_k c^\dagger_{\mathbf{k},\downarrow} c_{\mathbf{k},\downarrow} = \xi_k - \xi_k c_{\mathbf{k},\downarrow} c^\dagger_{\mathbf{k},\downarrow} \,.$$

Now we notice that in the Hamiltonian we have terms proportional to $cc, cc^\dagger, c^\dagger c, c^\dagger c^\dagger$, and such combinations can be expressed in terms of matrices of the form

TODO: put momentum and spin pedices

$$\begin{pmatrix} c^\dagger c & c^\dagger c^\dagger \\ cc & cc^\dagger \end{pmatrix} .$$

This algebraic maneuver allows us to recast the operator $\hat{K}$ into the following compact matrix form:

$$\hat{K} = \sum_{\mathbf{k}} \begin{pmatrix} c^\dagger_{\mathbf{k},\uparrow} & c_{-\mathbf{k},\downarrow} \end{pmatrix} \begin{pmatrix} \xi_k & -\Delta_{\mathbf{k}} \\ -\Delta^*_{\mathbf{k}} & -\xi_k \end{pmatrix} \begin{pmatrix} c_{\mathbf{k},\uparrow} \\ c^\dagger_{-\mathbf{k},\downarrow} \end{pmatrix} + C, \tag{2.3.2}$$

where the constant energy offset C is explicitly given by all the terms which do not depend on the fermionic operators:

$$C = \sum_{\mathbf{k}} \left(\xi_k + A^*_{\mathbf{k}} \Delta_{\mathbf{k}}\right) .$$

Notice that the off-diagonal terms in the matrix representation of $\hat{K}$ are proportional to the gap parameter $\Delta_{\mathbf{k}}$, which lets operators of opposite momenta couple to each other. This is a direct manifestation of the fundamental pairing mechanism that underlies the BCS theory of superconductivity. The constant term is important, but we want to "solve the problem" (diagonalize the Hamiltonian) in order to find the ground state and the quasiparticle excitations, without having to worry about the constant term, which will be added at the end.

Quasiparticle Transformation

Let us define the central 2×2 Hermitian matrix $M_{\mathbf{k}}$ as:

$$M_{\mathbf{k}} = \begin{pmatrix} \xi_k & -\Delta_{\mathbf{k}} \\ -\Delta^*_{\mathbf{k}} & -\xi_k \end{pmatrix} .$$

We now show that the many-body problem of diagonalizing the quadratic form $\hat{K}$ reduces entirely to the problem of diagonalizing the independent 2×2 matrix $M_{\mathbf{k}}$ for every momentum mode $\mathbf{k}$. In strict analogy to the procedure utilized for the weakly interacting Bose gas (with the Bogoliubov

transformation and quasiparticles), we seek a transformation to a new set of fermionic creation and destruction operators, $\gamma_{\mathbf{k},\sigma}$ and $\gamma^\dagger_{\mathbf{k},\sigma}$, that diagonalize $\hat{K}$. These will become our new *quasiparticle operators*. We introduce a transformation matrix $U_\mathbf{k}$ such that:

$$\begin{pmatrix} c_{\mathbf{k},\uparrow} \\ c^\dagger_{-\mathbf{k},\downarrow} \end{pmatrix} = U_\mathbf{k} \begin{pmatrix} \gamma_{\mathbf{k},\uparrow} \\ \gamma^\dagger_{-\mathbf{k},\downarrow} \end{pmatrix}, \quad \begin{pmatrix} c^\dagger_{\mathbf{k},\uparrow} & c_{-\mathbf{k},\downarrow} \end{pmatrix} = \begin{pmatrix} \gamma^\dagger_{\mathbf{k},\uparrow} & \gamma_{-\mathbf{k},\downarrow} \end{pmatrix} U^\dagger_\mathbf{k}. \tag{2.3.3}$$

Notice that we are mixing momenta and spins in this transformation, which is a direct consequence of the fundamental pairing mechanism that underlies the BCS theory (the structure of the Hamiltonian encodes this directly).

This transformation must simultaneously satisfy two critical conditions:

1. The terms appearing in the grand canonical Hamiltonian, once expressed in terms of the new operators, must form a diagonal matrix $D_\mathbf{k}$:

$$\begin{pmatrix} c^\dagger_{\mathbf{k},\uparrow} & c_{-\mathbf{k},\downarrow} \end{pmatrix} M_\mathbf{k} \begin{pmatrix} c_{\mathbf{k},\uparrow} \\ c^\dagger_{-\mathbf{k},\downarrow} \end{pmatrix} = \begin{pmatrix} \gamma^\dagger_{\mathbf{k},\uparrow} & \gamma_{-\mathbf{k},\downarrow} \end{pmatrix} U^\dagger_\mathbf{k} M_\mathbf{k} U_\mathbf{k} \begin{pmatrix} \gamma_{\mathbf{k},\uparrow} \\ \gamma^\dagger_{-\mathbf{k},\downarrow} \end{pmatrix},$$

   with $D_\mathbf{k} = U^\dagger_\mathbf{k} M_\mathbf{k} U_\mathbf{k}$ being explicitly diagonal: $U_\mathbf{k}$ is a unitary (as we will see in a moment) change of basis that diagonalizes the Hamiltonian.

2. The fundamental anti-commutation rules must be preserved for the new quasiparticle operators, ensuring in particular that:

$$\{\gamma_{\mathbf{k},\sigma}, \gamma^\dagger_{\mathbf{k}',\sigma'}\} = \delta_{\mathbf{k},\mathbf{k}'}\delta_{\sigma,\sigma'},$$

   with all remaining anti-commutators identically zero.

We can readily prove that requiring the transformation $U_\mathbf{k}$ to be unitary guarantees the preservation of these anti-commutation relations on the new operators.

Proof. Let us parameterize the inverse transformation matrix as:

$$U^{-1}_\mathbf{k} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}.$$

The relation linking the bare c operators to the quasiparticle γ operators is then:

$$\begin{pmatrix} \gamma_{\mathbf{k},\uparrow} \\ \gamma^\dagger_{-\mathbf{k},\downarrow} \end{pmatrix} = U^{-1}_\mathbf{k} \begin{pmatrix} c_{\mathbf{k},\uparrow} \\ c^\dagger_{-\mathbf{k},\downarrow} \end{pmatrix}, \quad \begin{pmatrix} \gamma^\dagger_{\mathbf{k},\uparrow} & \gamma_{-\mathbf{k},\downarrow} \end{pmatrix} = \begin{pmatrix} c^\dagger_{\mathbf{k},\uparrow} & c_{-\mathbf{k},\downarrow} \end{pmatrix} (U^{-1}_\mathbf{k})^\dagger,$$

where $(U^{-1}_\mathbf{k})^\dagger$ is the conjugate transpose of $U^{-1}_\mathbf{k}$:

$$(U^{-1}_\mathbf{k})^\dagger = \begin{pmatrix} a^*_{11} & a^*_{21} \\ a^*_{12} & a^*_{22} \end{pmatrix}.$$

TODO: check computations from Di Sante notes (Filippo)

By explicitly substituting this into the anti-commutators and using the known algebraic relations for the bare c operators, it is straightforward to check that:

$$\{\gamma_{\mathbf{k},\uparrow}, \gamma^\dagger_{\mathbf{k}',\uparrow}\} = \left\{ \left(a_{11}c_{\mathbf{k},\uparrow} + a_{12}c^\dagger_{-\mathbf{k},\downarrow}\right), \left(a^*_{11}c^*_{\mathbf{k}',\uparrow} + a^*_{12}c^\dagger_{-\mathbf{k}',\downarrow}\right) \right\} = (|a_{11}|^2 + |a_{12}|^2)\delta_{\mathbf{k},\mathbf{k}'},$$

$$\{\gamma_{-\mathbf{k},\downarrow}, \gamma^\dagger_{-\mathbf{k}',\downarrow}\} = \left\{ \left(a_{21}c_{-\mathbf{k},\downarrow} + a_{22}c^\dagger_{\mathbf{k},\uparrow}\right), \left(a^*_{21}c^*_{-\mathbf{k}',\downarrow} + a^*_{22}c^*_{\mathbf{k}',\uparrow}\right) \right\} = (|a_{21}|^2 + |a_{22}|^2)\delta_{\mathbf{k},\mathbf{k}'},$$

$$\{\gamma_{\mathbf{k},\uparrow}, \gamma_{-\mathbf{k}',\downarrow}\} = \left\{ \left(a_{11} c_{\mathbf{k},\uparrow} + a_{12} c^{\dagger}_{-\mathbf{k},\downarrow} \right), \left(a^*_{21} c^*_{-\mathbf{k}',\downarrow} + a^*_{22} c^*_{\mathbf{k}',\uparrow} \right) \right\} = (a_{11} a^*_{21} + a_{12} a^*_{22}) \delta_{\mathbf{k},-\mathbf{k}'}.$$

In order to successfully satisfy the fermionic algebra, the coefficients must obey:

$$1 = |a_{11}|^2 + |a_{12}|^2,$$

$$1 = |a_{21}|^2 + |a_{22}|^2,$$

$$0 = a_{11} a^*_{21} + a_{12} a^*_{22}.$$

These three constraints are automatically and exactly satisfied by unitary matrices. This implies that $U^{\dagger}_{\mathbf{k}} M_{\mathbf{k}} U_{\mathbf{k}} = U^{-1}_{\mathbf{k}} M_{\mathbf{k}} U_{\mathbf{k}}$; thus, finding a unitary matrix that diagonalizes the Hermitian matrix $M_{\mathbf{k}}$ will successfully diagonalize the entire many-body Hamiltonian $\hat{K}$. □

Quasiparticle Dispersion and Coherence Factors

The eigenvalues λ of $M_{\mathbf{k}}$ are determined by its characteristic equation:

$$(\xi_k - \lambda)(-\xi_k - \lambda) - |\Delta_{\mathbf{k}}|^2 = 0,$$

from which we immediately obtain the roots $\lambda = \pm E_{\mathbf{k}}$, where:

$$E_{\mathbf{k}} = \sqrt{\xi_k^2 + |\Delta_{\mathbf{k}}|^2} > 0. \tag{2.3.4}$$

This fundamentally important quantity $E_{\mathbf{k}}$ represents the *energy dispersion of the superconducting quasiparticle excitations*. We expected two opposite sign eigenvalues because the Hamiltonian ...

If we let the gap parameter $\Delta_{\mathbf{k}}$ vanish, the quasiparticle dispersion reduces to the free-particle kinetic energy relative to the chemical potential, $E_{\mathbf{k}} = |\xi_k|$. In this limit, the quasiparticles are simply the original bare fermions.

The normalized eigenvector of $M_{\mathbf{k}}$ corresponding to the positive eigenvalue $E_{\mathbf{k}}$ takes the form:

$$\begin{pmatrix} u^*_{\mathbf{k}} \\ -v^*_{\mathbf{k}} \end{pmatrix}, \text{ with } |u_{\mathbf{k}}|^2 + |v_{\mathbf{k}}|^2 = 1.$$

Following this definition, the **coherence factors** can be expressed as:

$$u^*_{\mathbf{k}} = \frac{\Delta_{\mathbf{k}}}{\sqrt{2E_{\mathbf{k}}(E_{\mathbf{k}} - \xi_k)}}, \quad v^*_{\mathbf{k}} = \frac{E_{\mathbf{k}} - \xi_k}{\sqrt{2E_{\mathbf{k}}(E_{\mathbf{k}} - \xi_k)}} = v_{\mathbf{k}},$$

such that $v_{\mathbf{k}}$ will be strictly real and positive, while $u_{\mathbf{k}}$ can be complex.

Conversely, the normalized eigenvector corresponding to the negative eigenvalue $-E_{\mathbf{k}}$ is:

$$\begin{pmatrix} v_{\mathbf{k}} \\ u_{\mathbf{k}} \end{pmatrix}.$$

Remark. *In defining these eigenvectors, a specific choice of phase factors has been made such that $v^*_{\mathbf{k}}$ is strictly real and the determinant of the transformation matrix is exactly* $\det U_{\mathbf{k}} = 1$.

Using these two orthogonal eigenvectors, one may immediately construct the explicit unitary transformation $U_{\mathbf{k}}$ that diagonalizes $M_{\mathbf{k}}$:

$$U_{\mathbf{k}} = \begin{pmatrix} u^*_{\mathbf{k}} & v_{\mathbf{k}} \\ -v^*_{\mathbf{k}} & u_{\mathbf{k}} \end{pmatrix},$$

with its corresponding inverse:

$$U^{-1}_{\mathbf{k}} = U^{\dagger}_{\mathbf{k}} = \begin{pmatrix} u_{\mathbf{k}} & -v_{\mathbf{k}} \\ v^*_{\mathbf{k}} & u^*_{\mathbf{k}} \end{pmatrix}.$$

Bogoliubov Transformation and Diagonalized Hamiltonian

By applying the previously derived inverse transformation, we obtain the explicit mathematical expressions for the new operators in terms of the old ones (the celebrated Bogoliubov transformation):

$$\gamma_{\mathbf{k},\uparrow} = u_{\mathbf{k}} c_{\mathbf{k},\uparrow} - v_{\mathbf{k}} c^{\dagger}_{-\mathbf{k},\downarrow}, \tag{2.3.5}$$

$$\gamma^{\dagger}_{-\mathbf{k},\downarrow} = v^{*}_{\mathbf{k}} c_{\mathbf{k},\uparrow} + u^{*}_{\mathbf{k}} c^{\dagger}_{-\mathbf{k},\downarrow}.$$

Notice that the new annihilation operator $\gamma_{\mathbf{k},\uparrow}$ physically mixes the old creation operator $c^{\dagger}_{-\mathbf{k},\downarrow}$ and destruction operator $c_{\mathbf{k},\uparrow}$, specifically entangling particles with opposite spin and momentum.[4]

In these transformed variables, spin plays a non trivial role, as the same operator creates a particle with spin up and destroys a particle with spin down. This is a direct consequence of the pairing mechanism, which couples opposite spins and momenta, preserving the *parity symmetry* Z_2 of the system. The Bogoliubov transformation is a canonical unitary transformation that preserves the fundamental anti-commutation relations.

Thinking about the *Stern-Gerlach experiment*, we can see that the new quasiparticle operators $\gamma_{\mathbf{k},\sigma}$ and $\gamma^{\dagger}_{\mathbf{k},\sigma}$ do not correspond to any physical spin measurement along a given axis. Instead, they represent a superposition of spin-up and spin-down states, which is a direct manifestation of the entanglement between opposite spins induced by the pairing mechanism of BCS theory.

Expressed in terms of these newly defined γ operators, the grand canonical Hamiltonian assumes its fully diagonalized form:

$$\hat{K} = \sum_{\mathbf{k}} \begin{pmatrix} \gamma^{\dagger}_{\mathbf{k},\uparrow} & \gamma_{-\mathbf{k},\downarrow} \end{pmatrix} D_{\mathbf{k}} \begin{pmatrix} \gamma_{\mathbf{k},\uparrow} \\ \gamma^{\dagger}_{-\mathbf{k},\downarrow} \end{pmatrix} + C,$$

where the diagonal eigenvalue matrix $D_{\mathbf{k}} = U^{\dagger}_{\mathbf{k}} M_{\mathbf{k}} U_{\mathbf{k}}$ is given by:

$$D_{\mathbf{k}} = \begin{pmatrix} E_{\mathbf{k}} & 0 \\ 0 & -E_{\mathbf{k}} \end{pmatrix}.$$

By explicitly expanding this matrix product and once again applying the quasiparticle anti-commutation relations, we obtain:

$$\hat{K} = \sum_{\mathbf{k}} \left(E_{\mathbf{k}} \gamma^{\dagger}_{\mathbf{k},\uparrow} \gamma_{\mathbf{k},\uparrow} - E_{\mathbf{k}} \gamma_{-\mathbf{k},\downarrow} \gamma^{\dagger}_{-\mathbf{k},\downarrow} \right) + C = \sum_{\mathbf{k},\sigma} E_{\mathbf{k}} \gamma^{\dagger}_{\mathbf{k},\sigma} \gamma_{\mathbf{k},\sigma} + C', \tag{2.3.6}$$

where the newly defined constant term C' absorbs the commutation shifts, assuring that the Hamiltonian is expressed in normal-ordered form:

$$C' = C - \sum_{\mathbf{k}} E_{\mathbf{k}} = \sum_{\mathbf{k}} \left(\xi_k + A^{*}_{\mathbf{k}} \Delta_{\mathbf{k}} - E_{\mathbf{k}} \right).$$

Because the quasiparticle energy dispersion $E_{\mathbf{k}}$ is strictly positive, the operator $\sum_{\mathbf{k},\sigma} E_{\mathbf{k}} \gamma^{\dagger}_{\mathbf{k},\sigma} \gamma_{\mathbf{k},\sigma}$ is positive semi-definite. Consequently, the constant C' physically represents the lowest possible eigenvalue of $\hat{K}$, which corresponds exactly to the ground state grand potential of the superconducting system.

[4] Equivalently, taking the Hermitian adjoint of the second equation yields $\gamma_{-\mathbf{k},\downarrow} = u_{\mathbf{k}} c_{-\mathbf{k},\downarrow} + v_{\mathbf{k}} c^{\dagger}_{\mathbf{k},\uparrow}$.

2.3.3 | Ground State Wave Function

The structure of the many-body wavefunction will change dramatically in the superconducting phase, considering also that the occupation of the quasiparticle states is now governed by the Fermi-Dirac distribution. But the ground state of the system is still defined as the state with the lowest possible energy, and can be easily obtained.

In the grand-canonical ensemble, the thermodynamic thermal average of any physical operator $\hat{O}$ is given by the standard statistical mechanics trace formula:

$$\langle O \rangle = \frac{\mathrm{tr}\left(e^{-\beta \hat{K}} \hat{O}\right)}{\mathcal{Z}},$$

where the grand partition function is defined as:

$$\mathcal{Z} = \mathrm{tr}\left(e^{-\beta \hat{K}}\right).$$

Here, $\hat{K} = \hat{H} - \mu \hat{N}$ represents the grand canonical Hamiltonian. In the absolute zero temperature limit, $T \to 0$, the inverse temperature $\beta \to +\infty$. In this asymptotic regime, the thermal average strictly reduces to a pure quantum mechanical expectation value over the ground state, where the grand-canonical ground state $|\psi_0(\mu)\rangle$ is defined as the eigenstate of $\hat{K}$ corresponding to its lowest possible eigenvalue[5].

Based on our preceding diagonalization procedure, the lowest energy state corresponds to the configuration devoid of any quasiparticle excitations (since the quasiparticle energy is strictly positive). Therefore, in the present case, $|\psi_0(\mu)\rangle$ is precisely the vacuum of the new Bogoliubov operators $\gamma_{\mathbf{k},\sigma}$, meaning that it must satisfy the annihilation condition:

$$\gamma_{\mathbf{k},\sigma} |\psi_0\rangle = 0 \quad \forall \mathbf{k}, \sigma. \tag{2.3.7}$$

By explicitly using the Bogoliubov transformation derived earlier to express $\gamma_{\mathbf{k},\uparrow}$ in terms of the bare particle operators,[6] this vacuum condition for the specific spin-up branch ($\sigma = \uparrow$) yields:

$$u_{\mathbf{k}} c_{\mathbf{k},\uparrow} |\psi_0\rangle = v_{\mathbf{k}} c^{\dagger}_{-\mathbf{k},\downarrow} |\psi_0\rangle. \tag{2.3.8}$$

Looking at (2.3.8), we can observe a profound physical symmetry: the action of destroying a bare fermion with momentum $\mathbf{k}$ and spin $\uparrow$ on the ground state is directly proportional to creating a bare fermion with opposite momentum $-\mathbf{k}$ and opposite spin $\downarrow$. This is a direct manifestation of Spontaneous Symmetry Breaking of the $U(1)$ gauge symmetry (conservation of particle number, broken in the ground state), which is a fundamental property of the superconducting phase. The ground state is a coherent superposition of states with different particle numbers, and the phase of the order parameter is spontaneously chosen.

This entangled behavior is structurally very similar to what we previously found for the correlated ground state of a dilute interacting Bose gas. In that bosonic case, we determined that the ground state wave function was heavily populated by pairs of opposite momenta and could be expressed using an exponential ansatz:

$$|\psi_0\rangle = A e^{B a^{\dagger}_{\mathbf{k}} a^{\dagger}_{-\mathbf{k}}} = A + \left(B a^{\dagger}_{\mathbf{k}} a^{\dagger}_{-\mathbf{k}}\right) + \frac{A}{2}\left(B a^{\dagger}_{\mathbf{k}} a^{\dagger}_{-\mathbf{k}}\right)^2 + \dots$$

[5] We are assuming that the ground state is unique. If this is not the case, a sum over all degenerate ground states would be required to evaluate the expectation values properly.

[6] For the sake of finding measurable observables, since the quasiparticles are a mathematical construct: we need correlation functions of the old fermionic variables, which can be measured.

with A and B representing specific coherent coefficients that we determined.

However, in the present scenario, we are strictly dealing with fermions. Fermions are governed by the Pauli exclusion principle, which mathematically enforces that the square of any fermionic creation operator identically vanishes ($c^\dagger c^\dagger = 0$). Consequently, all terms in the previous coherent expansion proportional to $\left(a^\dagger_{\mathbf{k}} a^\dagger_{-\mathbf{k}}\right)^n$ with $n > 1$ must vanish exactly for fermions.

Truncating the series to respect the Pauli principle, we thus try the following linear pairing ansatz for the wavefunction of a single momentum mode:

$$\left(A + B c^\dagger_{\mathbf{k},\uparrow} c^\dagger_{-\mathbf{k},\downarrow}\right)|0\rangle,$$

where $|0\rangle$ represents the absolute vacuum of the original bare fermionic operators $c_{\mathbf{k},\sigma}$ (meaning a state containing zero particles). To determine the unknown coefficients A and B, we strictly require that this proposed ansatz satisfies the defining vacuum condition (2.3.8).

Evaluating the left-hand side of the condition by acting with the destruction operator yields:

$$\begin{aligned} u_{\mathbf{k}} c_{\mathbf{k},\uparrow}\left(A + B c^\dagger_{\mathbf{k},\uparrow} c^\dagger_{-\mathbf{k},\downarrow}\right)|0\rangle &= B u_{\mathbf{k}} c_{\mathbf{k},\uparrow} c^\dagger_{\mathbf{k},\uparrow} c^\dagger_{-\mathbf{k},\downarrow}|0\rangle \\ &= B u_{\mathbf{k}} c^\dagger_{-\mathbf{k},\downarrow}|0\rangle. \end{aligned}$$

Evaluating the right-hand side of the condition by acting with the creation operator yields:

$$v_{\mathbf{k}} c^\dagger_{-\mathbf{k},\downarrow}\left(A + B c^\dagger_{\mathbf{k},\uparrow} c^\dagger_{-\mathbf{k},\downarrow}\right)|0\rangle = A v_{\mathbf{k}} c^\dagger_{-\mathbf{k},\downarrow}|0\rangle.$$

By comparing these two results, we immediately see that the proposed state successfully verifies the vacuum condition if and only if $B = v_{\mathbf{k}}$ and $A = u_{\mathbf{k}}$. Because the different momentum modes $\mathbf{k}$ are decoupled in the diagonalized Hamiltonian, this structural requirement must be independently satisfied for every momentum pair $(\mathbf{k}\uparrow, -\mathbf{k}\downarrow)$ across the entire system. Taking the tensor product over all possible momentum states yields the complete many-body ground state:

$$|\psi_0(\mu)\rangle = \prod_{\mathbf{k}}\left(u_{\mathbf{k}} + v_{\mathbf{k}} c^\dagger_{\mathbf{k},\uparrow} c^\dagger_{-\mathbf{k},\downarrow}\right)|0\rangle. \tag{2.3.9}$$

The state described by Equation 2.3.9 is the celebrated BCS wave function. It was first proposed historically by Bardeen, Cooper, and Schrieffer in 1957 as a brilliant variational wave function to approximate the ground state of a metallic superconductor. The macroscopic state describes a coherent superposition where each momentum pair state $(\mathbf{k}\uparrow, -\mathbf{k}\downarrow)$ is either empty with probability amplitude $u_{\mathbf{k}}$, or occupied by a Cooper pair with probability amplitude $v_{\mathbf{k}}$.

Finally, one can easily check that this constructed state is properly normalized to unity. Indeed, because different momentum modes commute, the norm factorizes as:

$$\langle\psi_0|\psi_0\rangle = \prod_{\mathbf{k}}\left(|u_{\mathbf{k}}|^2 + |v_{\mathbf{k}}|^2\right) = 1,$$

which is guaranteed to equal 1 thanks to the fundamental normalization condition on the coherence factors, $|u_{\mathbf{k}}|^2 + |v_{\mathbf{k}}|^2 = 1$, that we rigorously enforced during the Bogoliubov unitary transformation.

Exercises

Sheet 1

[...]

Appendices

A | Notation and Conventions

B | Computations and further developments